Arbitrage Opportunities and Efficiency Tests in Crypto Derivatives

Carol Alexander Xi Chen Jun Deng Tianyi Wang

Highlights

1. We formulate a novel fiat-currency-free put-call parity relationship.
2. These form the basis of several efficiency tests for crypto derivatives markets.
3. Bitcoin markets are more efficient than ether.
4. Arbitrage opportunities in ether derivatives are particularly profitable.
5. Determinants of arbitrage profits are investigated.

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Arbitrage Opportunities and Efficiency Tests in Crypto Derivatives

Carol Alexander[a,b,d] Xi Chen[a] Jun Deng[c] Tianyi Wang[c]

Abstract

We test the joint efficiency of the bitcoin and ether options and perpetual futures markets and identify the determinants of arbitrage opportunities. Our novel fiat-currency-free put-call parity relationship motivates new arbitrage tests for options-only and option-perpetual cross-markets. Bitcoin and ether derivatives markets are becoming more efficient, especially for options of maturity ≥ 15 days. Bitcoin derivative markets are generally more efficient than ether derivative markets, but arbitrage strategies can still be highly profitable even under conservative transaction cost scenarios, which include slippage for large orders, especially during periods of high trading volumes or when the blockchain traffic becomes more congested.

Keywords: Box Spread; Calendar Spread; Inverse Option; Put-Call Parity; Transaction Costs

JEL Classification: G11; G32

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Data declaration: We thank CoinMetrics for providing the Deribit options and futures tick-by-tick data set.

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1 Introduction

In January 2020 the Chicago Mercantile Exchange (CME) began listing bitcoin options written on its bitcoin futures contracts. Being the only regulated crypto derivatives trading platform and one that is geared towards large institutional clients, this marked a turning point in crypto markets. Crypto markets used to be highly segmented with extreme volatility caused by excessive price manipulation. So large asset managers were unwilling to invest. They were, theoretically, unable do so anyway, due to U.S. regulations.[1] However, when the CME listed bitcoin and ether options these institutions became able to hedge volatility as well as price risk, and they began to trade in the only regulated crypto derivatives market, on the CME. Institutional investment in crypto is a real game changer, as we have recently witnessed with the huge inflows to Blackrock and other bitcoin spot exchange-traded funds (ETFs). Shortly after the ETFs were listed on the New York Stock Exchange the price of bitcoin almost doubled in the space of a few weeks, reaching a new all time high of $73,000.[2]

More dominant (but unregulated) crypto exchanges, especially Deribit but also BitMEX, Binance, OKEx and the now defunct FTX.US (Ledger-X) had already been providing trading platforms for bitcoin futures and options several years earlier, but it was not until the end of 2021 that liquidity in bitcoin options became sufficient for high-frequency trading. At that time the Dutch exchange Deribit, founded in 2016, relocated to Panama and subsequently became the main venue not only for bitcoin but also for trading ether options, having about 85% of the trading volumes and 90% of open interest at the time of writing. The unregulated Deribit platform facilities much higher leverage than CME and has deeper and more liquid trading, which attracts a broader scope of market participants, including both retail and institutional clients, as well as many professional speculative traders.

Similar to traditional options, the crypto options market is predominantly fueled by the hedging and speculative demands of crypto investors. A large extant literature on traditional financial options market establishes their role in facilitating the flow of information about the underlying asset (Chan et al., 2002; Carr and Wu, 2009; Jin et al., 2012). For speculators, Alexander et al. (2023b)

[1] Nevertheless, the Securities and Exchange Commission (SEC) has stated that the use of off-shore accounts or virtual private networks (VPNs) has still allowed such investments. See, for instance, the successful Commodity and Futures Exchange Commission and Department of Justice civil actions against Binance in November 2023, Release 8825-23.

[2] Even though the only really actively-traded options are highly concentrated both in terms of underlying (bitcoin and ether) and trading venue (the Deribit platform), other token prices are very highly correlated with these two particular tokens (Gkillas and Siriopoulos, 2023; Katsiampa, 2019).

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found that bitcoin options play a unique role in enabling both volatility-informed and directional-informed trades.[3] For hedgers, options are even more important in crypto than in traditional markets, because of the high volatility of crypto prices.[4] But, unlike traditional options, the Deribit products are quoted and settled in crypto, which allows trading to be entirely fiat-currency-free. Furthermore, the spot and futures markets are still highly segmented across multiple platforms and thus exhibit recurrent cross-market arbitrages (Makarov and Schoar, 2020).

On 10 January 2024 the U.S. Securities and Exchange Commission (SEC) eventually approved eleven bitcoin spot ETF applications.[5] Indeed, at the time of writing (March 2024) theblock.co already reports about $60 billion assets under management in bitcoin spot ETFs. The asset managers that offer these bitcoin ETFs to their clients need to hedge both price and volatility risk, and they will be doing the latter using Deribit bitcoin options. Obviously, such large institutional players would benefit from improved efficiency in these markets. That is why this paper poses these new questions about the efficiency of crypto derivatives markets, and links the profitability of arbitrage strategies to the mix of professional, informed traders with retail, less-informed traders.

We investigate the efficiency of the Deribit bitcoin and ether options markets, because over 85% of total market volume across all option exchanges is currently traded on Deribit.[6] We also test the joint efficiency of the option-perpetual markets because the perpetual futures is the favoured instrument for delta hedging and has much greater volumes than calendar futures as a result. There is already some very strong empirical evidence supporting the role of perpetuals in both the price discovery process and hedge efficiency of the bitcoin market.[7] One unique aspect of our study is to assess the efficiency of the option-perpetual cross market using a new, model-free and fiat-currency-free put-call parity relationship.

Bitcoin and ether together now account for over 70% of the total crypto market capitalization,

[3]Even though the options market is relatively new, other research shows that these markets are rich in information about the future. For instance, Hoang and Baur (2020) shows that Deribit bitcoin implied volatility contains unique information about forecasting short-term spot volatility. Moreover, bitcoin's realised volatility, volatility index, and variance risk premium are correlated with their equivalent for U.S. equities, oil, gold, the USD/EUR exchange rate, and the 10-year U.S. Treasury note (Alexander and Imeraj, 2021) and the bitcoin option implied volatility surface is mainly driven by volatility-informed traders (see Alexander et al., 2023b; Hou et al., 2020; Cao and Celik, 2021; Siu and Elliott, 2021, for research of bitcoin option pricing).

[4]See for instance Scaillet et al. (2020) and Kolokolov (2020) for evidence of frequent price jumps in the bitcoin spot price.

[5]These are: ARK 21Shares Bitcoin ETF (ARKB), Bitwise Bitcoin ETF (BITB), Fidelity Wise Origin Bitcoin Trust (FBTC), Franklin Bitcoin ETF (EZBC), Grayscale Bitcoin Trust (GBTC), Hashdex Bitcoin ETF (DEFI), Invesco Galaxy Bitcoin ETF (BTCO), iShares Bitcoin Trust (IBIT), Valkyrie Bitcoin Fund (BRRR), VanEck Bitcoin Trust (HODL), and WisdomTree Bitcoin Fund (BTCW).

[6]As yet there are no cross-asset (e.g. spread) options listed, so we cannot test for the joint efficiency of bitcoin and ether options.

[7]See for instance Urquhart (2016); Corbet et al. (2018); Kapar and Olmo (2019); Alexander et al. (2020b, 2023c).

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as of March 2024. However, almost all the previous literature on crypto derivatives focuses only on bitcoin. Few papers analyze ether derivatives,[8] but during the past two years the trading volume on ether and its derivatives has surged, so investigating ether now becomes as crucial as analysing bitcoin.

Our original theoretical contribution is to derive fiat-currency-free efficiency tests that can be applied to crypto derivatives markets. More specifically, we reformulate the classic put-call parity (PCP) relationship to a fiat-currency-free PCP. This is necessary because Deribit is a fiat-free exchange, meaning that its crypto derivatives are settled in crypto (as aforementioned) even though the underlying is quoted in fiat currency against the crypto.[9] The fiat-currency-free PCP yields a test for no-arbitrage violations of the cross option-perpetual market and a similar box-spread test is derived for the options market alone. Each test accounts for the transactions cost involved in forming the arbitrage portfolio. The frequency with which market prices violate these arbitrage price boundaries yields a direct indication of how often the market becomes inefficient with respect to the no-arbitrage condition. Then, by assuming the arbitrage price boundaries always hold, we extract the implied transaction costs on options and perpetuals from their market prices. The higher these implied costs, the lower the degree of efficiency in these markets.

Subsequently, we apply our efficiency tests to a very rich data set on Deribit perpetuals and options, showing how market efficiency has been improving over time, not only for the bitcoin and ether options markets alone but also (albeit to a lesser extent) for the option-perpetual cross markets. We also analyze option market efficiency by maturity, finding that the greatest efficiency improvements have been in longer-dated options (≥ 15 days). Bitcoin markets are still more efficient than ether derivatives markets, potentially because the ether markets are still too young. In fact, these two markets have displayed different characteristics over time: for bitcoin, the options market is significantly more efficient than the option-perpetual cross market but this is not so for ether; both the ether markets are continually evolving towards a more efficient state, while the bitcoin derivatives market lost efficiency in 2021 due to some extreme moves in perpetual prices in April, May and July of that year.

Finally, following the ideas first suggested by Crépellière et al. (2023) for crypto spot markets,

[8] But see price discovery in ether markets, Alexander et al. (2020a) and Brauneis and Mestel (2018).

[9] The inverse payoff structure induces price and volatility features significantly different from their traditional counterparts (Deng et al., 2021, 2020; Alexander et al., 2023a). In particular it is easier to hedge inverse options using inverse futures. One can also consider direct futures, which are like standard futures only versus tether (USDT) or another stable coin, rather than USD, but then one would need to include a dynamic hedge of the margin account. Furthermore, for delta hedging one could dynamically trade the bitcoin inverse perpetual instead of the underlying spot since the basis risk is minuscule.

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we investigate the determinants of arbitrage in options and perpetuals markets. To this end we construct several potential explanatory variables, these being divided into four different categories, namely perpetual specific, option specific, blockchain specific and market wide. Since the implied transactions costs are inversely related to market efficiency, higher implied transaction costs also indicate greater arbitrage profits. Unfortunately, with so many potential determinants, multicollinearity between some of the significant factors could become an issue, so our final regression is a result of a careful paring down keeping this multicollinearity in mind. In the end, after an extensive empirical investigation, we pin down only a few uncorrelated drivers of arbitrage in bitcoin and ether options and perpetuals markets. One of the main drivers of arbitrage in all markets (and especially the bitcoin option market) is the trading volume on perpetuals, which is highly correlated with other variables such as Google searches. One possible explanation for this is that informed, high-frequency traders are making arbitrage profits at the expense of less-informed retail traders. Due to the lack of regulation in these markets, a main causal factor of arbitrage opportunities could be the prevalence of algorithmic traders – especially during periods of high volatility (of which they themselves may in fact be the cause). Arbitrage profits in the ether options market are also impacted by blockchain specific variables such as gas fees and the network activity.

In the following: Section 2 provides background information on the markets and formulates the derivative payoffs to facilitate further discussions; Section 3 revises the traditional PCP to a fiat-currency-free version and constructs our efficiency tests, including a box-spread test for efficiency of the options market alone. This section also derives implied transaction costs assuming no violation of the arbitrage boundaries derived from the PCP and box-spread tests; Section 4 introduces the data required for our analysis, including our trade matching process and our assumptions about transactions costs; Section 5 reports our empirical findings about the efficiency of the options and option-perpetual markets and addresses the determinants of arbitrages in bitcoin and ether derivatives markets. Section 6 concludes. An appendix describes the explanatory variables used for our analysis of the determinants of arbitrage, and derives further tests for weak-form arbitrage. All algorithms are written in Python and Matlab, and these are available upon request.

2 Background

Bitcoin has a market capitalization of around 1.25 trillion USD at the time of writing (March. 2024) representing a 53% share of the overall crypto market, and ether has a market capitalization of

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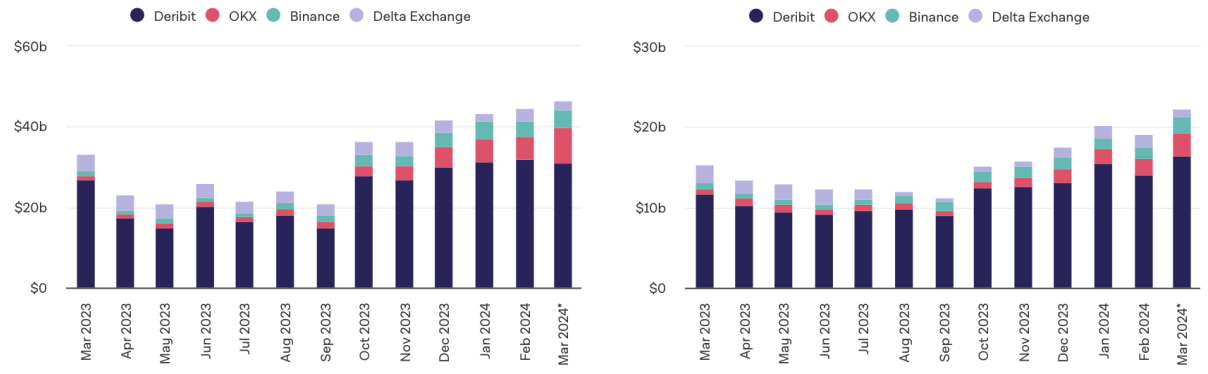
about 412 billion USD, or roughly 17.5%. Most trading volume is focused not on spot, or calendar futures, but on the perpetuals.[10] For instance, on the Deribit platform the daily trading volume of perpetuals is around 200 million USD, which is about 95% of the total volume over all the futures listed on Deribit.[11] The dominant position of Deribit in crypto option trading is reflected by its trading volume, whose evolution over the last twelve months, along with its disaggregation into four different platforms, is depicted in Figure 1. The figure was drawn on 23 March 2024, when the open interest on bitcoin options aggregated over these four platforms was approximately $20bn with 90% of these contracts traded on Deribit. The open interest on ether options aggregated over these four platforms was approximately $10bn, again with about 90% of these contracts trading on Deribit.

Figure 1: Deribit Monthly Options Trading Volumes The share of trading volume of bitcoin options (left) and ether options (right) from leading option exchanges, downloaded from https://www.theblock.co on 23 March 2024.



The exchange operates 24/7 and specializes in options but also lists the associated perpetual and calendar futures, which are mainly used for delta hedging the options (Alexander et al., 2023c). The underlying reference price for the bitcoin (ether) options is the Deribit bitcoin (ether) index, which is calculated every six seconds using an average of the four remaining bid-ask prices from six leading online crypto exchanges: Bitfinex, Gemini, Bitstamp, GDAX, Kraken, and Itbit. This approach helps to mitigate the risk of flash-crashes or short squeezes affecting any individual exchange.[12] The platform lists quarterly bitcoin (ether) futures contracts running on a March, June, September, December cycle and offers leverage of up to 100X, selected by the client, with cash settlement in BTC and a nominal value of $10 USD ($1 USD) per contract. The options are European-type

[10] See discussion in Alexander et al. (2023a) and the CryptoCompare Monthly Review.

[11] However, the main markets for perpetuals are other exchanges like Binance, Bybit, OKX and BitMex – but none of these are important platforms for options trading.

[12] For more details on market micro-structure and trading arrangements, refer to the Deribit documentation.

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with maturities ranging from daily, two-day, weekly, bi-weekly, to monthly, quarterly, and yearly expiration. This allows investors to engage in short-term speculation, hedge against jump risk, and manage long-run risk. The options are priced in BTC (ETH), but USD quotes are also provided through synthetic futures contracts with the same maturity as the corresponding options.[13]

To understand these contracts, let us use bitcoin (BTC) as an example; equivalent identities for ether (ETH) options are obvious. Consider a vanilla call option, with price $C_t(K,T)$ at time t, which gives the holder the right to buy one bitcoin at time T with the strike price of K. The option prices are measured as a percentage of one bitcoin – for instance, $C_t(K,T) = 10\%$ for an at-the-money option, but option strikes are quoted in USD, as are the underlying bitcoin price index S_t, $t \in [0,T]$ and the inverse futures prices $F_{t,T}$.

In the absence of transactions costs, the bitcoin call option has notional $N^{\mathbb{B}}$ measured in BTC, because the fiat-free Deribit platform settles the options in bitcoin, and the expiry payoff takes the form:

$$V_C^{\mathbb{B}}(K,T) = N^{\mathbb{B}} \cdot \frac{\max\left(S_T - K, 0\right)}{S_T} = N^{\mathbb{B}} \cdot \max\left(1 - \frac{K}{S_T}, 0\right). \tag{1}$$

Similarly, the payoff to a bitcoin put option, with price $P_t(K,T)$, at time T is:

$$V_P^{\mathbb{B}}(K,T) = N^{\mathbb{B}} \cdot \frac{\max\left(K - S_T, 0\right)}{S_T} = N^{\mathbb{B}} \cdot \max\left(\frac{K}{S_T} - 1, 0\right). \tag{2}$$

Unlike traditional vanilla European options that have payoff $\max(S_T - K, 0)$, which is a piece-wise linear function of S_T, bitcoin options have a payoff that is a function of $1/S_T$. That is, the options payoff structures exhibit an inverse relationship with the USD price of the crypto, hence the term *inverse options*. Figures 2a and 2b further illustrate how the inverse feature undermines the piece-wise linearity of the option's terminal payoffs related to S_T. This is particularly relevant to our construction of no-arbitrage portfolios in the next section, as the inverse feature of options means that one can most easily hedge them using inverse crypto futures, which we discuss next.

Similar to crypto options, the inverse futures also have payoff inversely related to the USD price of the crypto. To see this, let us again take bitcoin as an example. The futures contracts are in USD-denominated notional amount $N^{\$}$. At expiry time T, the payoff, denominated in BTC, to a

[13] There are also end-of-month and end-of-week futures, although they are less liquid than the other quarterly futures and much less liquid than the perpetual.

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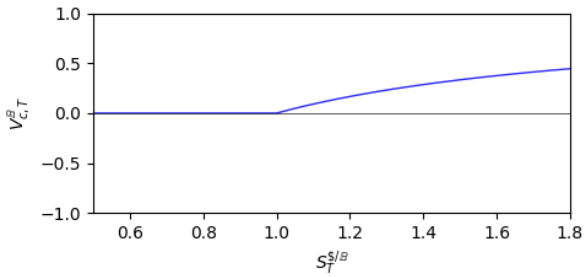
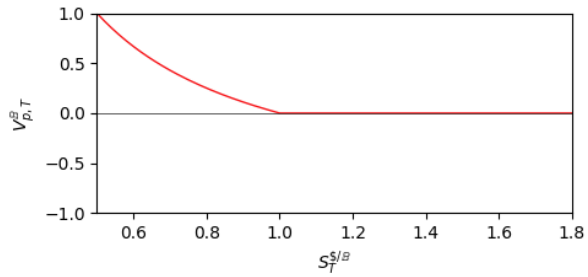
Figure 2: Expiry Payoff to Bitcoin Options and Futures, in BTC. The vertical axis shows the payoff in BTC, the horizontal axis is the underlying bitcoin index price S_T measured in USD. For simplicity we assume the options are bought at-the-money and the current bitcoin index price is $S_0 = 1$.

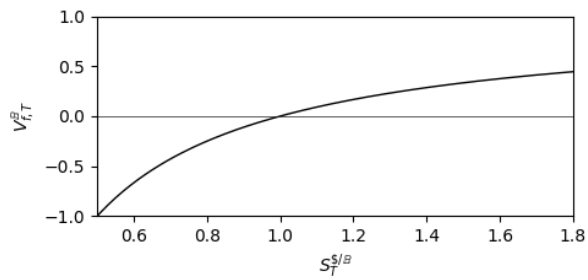
(a) The payoff of a bitcoin call option, $V^{\mathbb{B}}_{c,T}$.



(b) The payoff of a bitcoin put option, $V^{\mathbb{B}}_{p,T}$.



(c) The payoff of a bitcoin inverse futures contract, $V^{\mathbb{B}}_{f,T}$.



long position on the inverse futures takes the form:

$$V^{\mathbb{B}}_{t,T} = N^{\$}\left(\frac{1}{F_{t,T}} - \frac{1}{S_T}\right).$$

Like inverse options, bitcoin inverse futures have payoff measured in BTC, which is a function of $1/S_T$ as seen in Figure 2c.

Deribit provides a detailed list of transaction cost items for trading the inverse options and futures and these are summarized in Table 1.[14] Transaction costs are a key component of shaping market efficiency, and its evolution. Microstructure features influence users' participation on the blockchain; for instance, transaction costs can shift bitcoin miners from a mining-based to a market-based strategy (Easley et al., 2019). We highlight that the nominal size of an options contract is in BTC (ETH) but the futures contract has a nominal denominated in USD. Therefore transaction costs on options contracts are fixed, per contract in BTC (ETH). But for futures there is a fixed percentage transaction cost per contract, so fees are fixed in USD and consequently floating in BTC (ETH). This is particularly relevant to our empirical findings later on where we find that the options market has stable transaction costs in BTC (ETH), implied by the option prices, whereas the implied costs on futures contracts can be highly correlated with the USD prices of BTC (ETH).

[14] More details can be found on https://www.deribit.com/pages/docs/futures.

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In the discussion of costs in Section 4.4 we consider aggregate transaction costs, which include all the transaction cost items listed in Table 1 as well as trading costs from large orders due to price slippage.

Table 1: Deribit Option and Futures Transaction Fee Arrangements A "maker" order adds liquidity to the order book, which is not executed instantly, rather placed in the order book. A "taker" order takes liquidity from the order book, and it is executed instantly against other orders upon submission to the exchange. The withdrawal fee depends on the current state of the bitcoin network. More details are available at: Deribit.

Fee per Contract	Bitcoin (Ether) Futures	Bitcoin (Ether) Options
Maker/Taker Fee	0.00%/0.05% of the underlying	0.03% of the underlying or 0.0003 BTC (ETH) per contract
Delivery Fee	0.025%	0.015%
Withdrawal Fee	Varies	Varies
Deposit Fee	Free	Free

3 Testing the Efficiency of Crypto Derivatives Markets

This section proposes a wide rage of empirical tests on the efficiency of an options market, and the integrated option-perpetual cross market, as modified versions of existing efficiency tests for traditional derivatives markets (see for instance Black and Scholes, 1972; Ross, 1976; Chiras and Manaster, 1978). More specifically, we first revise the traditional PCP relationship, derived from the no-arbitrage principle using traditional call and put options, and futures contracts. Accommodating the inverse features of crypto options and futures contracts, we formulate a fiat-currency-free PCP for crypto derivatives. Using this PCP relationship the cross-market efficiency can be assessed in a model-free manner.[15]

Based on the revised PCP, we derive empirical tests for the efficiency of the option-perpetual cross market and the options market alone. Moreover, the tests include the aggregate transaction costs when trading the arbitrage portfolio(s). Failing a test exhibits arbitrage opportunities, in other words, it indicates a lack of efficiency in the market.[16]

In addition to the efficiency tests, we back out the aggregate transaction cost of trading arbitrage portfolios implied in both the options market and the option-perpetual market, assuming our

[15]Notable examples of this approach include studies by Ackert and Tian (2001); Klemkosky and Resnick (1979); Cremers and Weinbaum (2010) that investigate the mispricing and efficiency of the S&P 500 index options market.

[16]More tests are derived and included in the Appendix, Section B. Compared to the tests in this section, they are loosely derived. Passing them does not necessarily mean that the cross market is efficient. By contrast, the tests presented in this section are strictly derived from the PCP. Therefore passing these tests is an *absolute* indicator of an efficient market, because it means there are no arbitrage opportunities in the market regardless of traders' behavior or market price dynamics (Stoll, 1969).

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arbitrage relationship holds with the transaction costs included. Empirically, these implied transaction costs reveal the level of market efficiency through a different angle, compared to counting the frequency of the derivatives prices failing the efficiency tests. While the failure rate informs us how frequently arbitrage opportunities appear in the market, the implied transaction cost informs us to what extent the options market and option-perpetual cross market lack efficiency: the higher are these costs the less efficient are the markets and the greater the arbitrage gains. Combining our findings on the implied transaction costs and the failure rates of passing the efficiency tests, we may observe episodes when there are frequent arbitrage opportunities (reflected by high failure rates of efficiency tests), with low arbitrage gains (reflected by lower implied transaction costs); or infrequent but very profitable arbitrage opportunities.

3.1 The Fiat-Currency-Free PCP

We start by assuming no transaction costs, and a risk-free rate r for borrowing or lending crypto, following the derivation of the conventional PCP. We consider the following portfolio, constructed at time $0 \leq t \leq T$ using bitcoin options and futures – both settled in BTC:

- *Buy one call and write one put with the same strike price K and maturity T.*
  Value at time t is $C_t(K,T) - P_t(K,T)$ and payoff at time T is $1 - \frac{K}{S_T}$ per bitcoin;
- *Borrow $(1 - K/F_{t,T})e^{-r(T-t)}$ BTC at rate r.*
  Value at time t is $-\left(1 - \frac{K}{F_{t,T}}\right) e^{-r(T-t)}$ and payoff at time T is $-\left(1 - \frac{K}{F_{t,T}}\right)$ per bitcoin;
- *Short $K/N^{\$}$ units of bitcoin futures contracts that have a nominal size $N^{\$}$ and mature at T.*
  Value at time t is 0 and payoff at time T is $\frac{K}{S_T} - \frac{K}{F_{t,T}}$ per bitcoin.

The sum of the time T values is zero. If there is no arbitrage in the market, the portfolio should always yield a total payoff of 0 at time T and so would not cost anything at time t, so the sum of the three time t values above should be zero, viz.:

$$C_t(K,T) - P_t(K,T) - \left(1 - \frac{K}{F_{t,T}}\right) e^{-r(T-t)} = 0. \qquad \text{[PCP*]} \tag{3}$$

In a friction-less market the PCP relationship (3) should hold perfectly. But now we add transaction costs to the PCP portfolio, denoting by tc_o and tc_f the aggregate transaction costs, as a percentage of the option and futures contract size, respectively.[17]

[17] This follows the convention of transaction costs provided by Deribit, see Table 1.

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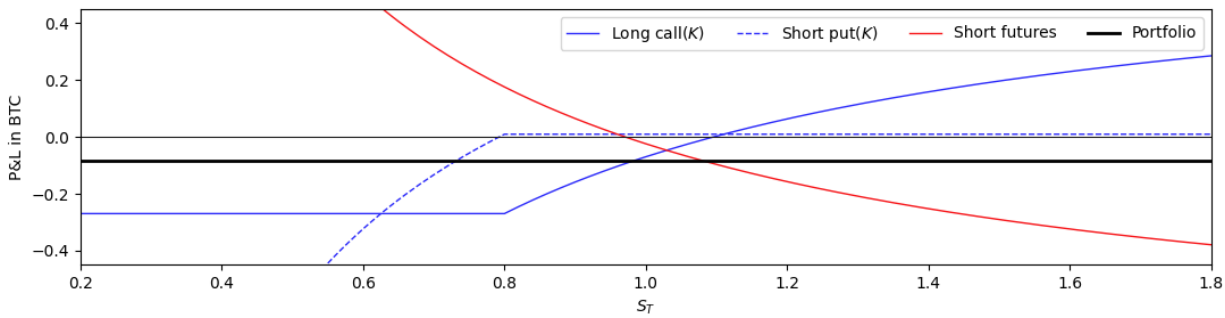
The transaction costs of constructing the PCP portfolio consist of those on:

(a) Two options whose cost in BTC is $2tc_o$ per bitcoin; and

(b) $K/N^{\$}$ units of futures contracts whose BTC value per contract at time t is $N^{\$}/F_{t,T}$ so the transaction cost in BTC is $(K/N^{\$})(N^{\$}/F_{t,T})\,tc_f = (K/F_{t,T})\,tc_f$.

Assuming the PCP relationship (3) holds, the arbitrage portfolio in BTC would always make a loss equal to the aggregate transaction cost of forming the portfolio, i.e. $2tc_o + (K/F_{t,T})tc_f$; see Figure 3 for an illustration.

Figure 3: PCP Portfolio P&L in BTC versus BTC Price S_T. Parameters for establishing the portfolio at time t: $F_{t,T} = S_t = 1$, $K = 1$, $C_t(K,T) = 0.2397$, $P_t(K,T) = 0.0397$, $r = 0$, and $tc_o = tc_f = 0.03$ per contract so the transaction costs on two options and K/N_f units of futures are $2tc_o + (K/F_{t,T})tc_f = 0.084$. The portfolio value at time t is $C_t(K,T) - P_t(K,T) - (1 - K/F_{t,T}) = 0$ and the portfolio payoff at time T is 0. So the portfolio P&L equates the total transaction costs, illustrated by the small and constant distance from the thick black line to zero. The prices and transaction costs are set at unrealistic levels only for graphing reasons.



When the derivative prices deviate from the PCP relationship (3), the time t value (and consequently the P&L) of the portfolio can move away from the aggregate value of the transaction costs. But there is no arbitrage opportunity when the profit generated by buying or selling the arbitrage portfolio together is less than the cost of forming it, i.e. when:

$$\left|C_t(K,T) - P_t(K,T) - \left(1 - \frac{K}{F_{t,T}}\right) e^{-r(T-t)}\right| \leq 2\,tc_o + \frac{K}{F_{t,T}} tc_f\,, \quad \text{[PCP test]} \tag{4}$$

Any violation of (4) would then exhibit a profitable arbitrage opportunity. One can make profits by buying (selling) calls, selling (buying) puts, hedging with an appropriate nominal of inverse futures and locking-in the profit, with a bitcoin borrow (lending) rate of r.[18] Violation of the PCP test is therefore evidence of inefficiency in the joint options-futures market. Thus, calibrating against (4) forms our first market efficiency test with the following null and alternative hypotheses:

H0. The payoff to the PCP portfolio is less than the cost of the portfolio, i.e. (4) holds and so

[18] See https://bitcompare.net/coins/bitcoin.

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the options-futures market is efficient;

H1. The payoff to the PCP portfolio exceeds the cost of the portfolio, i.e. (4) is violated and so the options-futures market lacks efficiency.

We remark that, since both options and futures are involved, the PCP test (4) is a joint test of the option-futures cross market. Violations of the PCP test could stem from the options and/or the futures market. In the following, we propose a box-spread test for the options market alone.

3.2 The Box-Spread Test for Option Market Efficiency

To test the efficiency of the options market alone we derive the arbitrage boundaries of the box spread using crypto options, following Ackert and Tian (2001). In a friction-less market, we could buy K_2 units of the PCP portfolio of (3) with strike K_1 and sell K_1 units of a different PCP portfolio of (3) with the same maturity but a different strike K_2, with $K_2 > K_1$. In this way, the new portfolio – a box spread, involves no futures contracts and should satisfy the equality derived from the difference between two PCPs of equation (3) for $K = K_1, K_2$, viz., writing $C_t(K_1, T) := C_{1t}$, $P_t(K_1, T) := P_{1t}$, $C_t(K_2, T) := C_{2t}$ and $P_t(K_2, T) = P_{2t}$ for brevity:

$$K_2(C_{1t} - P_{1t}) - K_1(C_{2t} - P_{2t}) - e^{-r(T-t)}(K_2 - K_1) = 0, \quad \text{[Box Spread]}$$

The transaction costs of forming the box spread are those on:

(a) K_2 units of call and put options with a strike K_1, whose cost in BTC is $2K_2 tc_o$; and

(b) K_1 units of call and put options with a strike K_2, whose cost in BTC is $2K_1 tc_o$.

The P&L of the box-spread portfolio should be equal to the transaction costs of forming the portfolio, $2tc_o(K_1 + K_2)$; see Figure 5 for an illustration.

Empirically, there will be no arbitrage opportunity in the market provided the profit generated by the assets in the portfolio together does not exceed the cost of forming the portfolio, which is $2tc_o(K_1 + K_2)$. In other words:

$$\left| K_2(C_{1t} - P_{1t}) - K_1(C_{2t} - P_{2t}) - e^{-r(T-t)}(K_2 - K_1) \right| \leq 2tc_o(K_1 + K_2), \quad \text{[box-spread test]} \quad (5)$$

where (C_{1t}, P_{1t}) and (C_{2t}, P_{2t}) are two pairs of options that are traded at the same time, with the same time to maturity, but different strikes: K_1 for (C_{1t}, P_{1t}) and K_2 for (C_{2t}, P_{2t}). The inequality (5) forms the basis of our second test for market efficiency, with the following null and alternative

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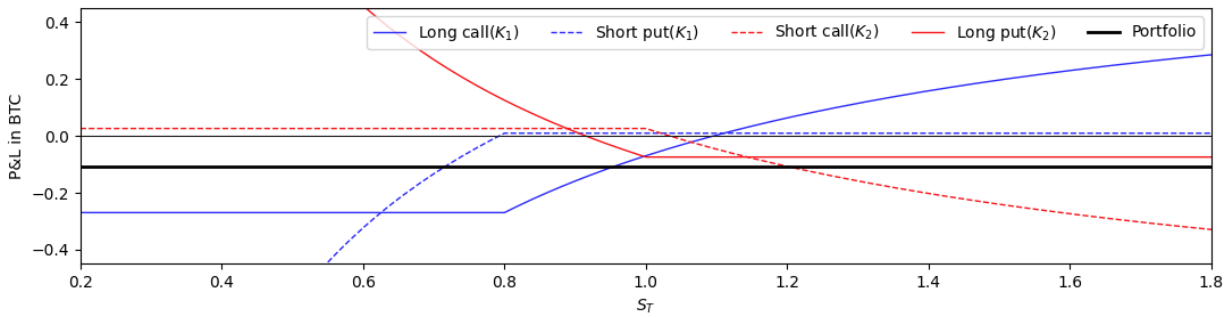
Figure 4: Box-Spread Portfolio P&L in BTC against the Bitcoin Price at Time T, S_T. Parameters for establishing the portfolio at time t: $F_{t,T} = S_t = 1$, $K_1 = 0.8$, $K_2 = 1$, $C_{1t} = 0.2397$, $P_{1t} = 0.0397$, $C_{2t} = 0.0628$ and $P_{2t} = 0.0628$, $tc_o = tc_f = 0.03$ per contract, $r = 0$. The crypto spot and futures prices and derivatives transaction costs are set at unrealistic levels only for graphing reasons. The portfolio value at time t is $K_2(C_{1t} - P_{1t}) - K_1(C_{2t} - P_{2t}) - (K_2 - K_1) = 0$ with the transaction cost on $2K_2 + 2K_1$ units of options, i.e. $(2K_2 + 2K_1)tc_o = 0.108$. The portfolio payoff at time T is 0 and so, the portfolio P&L equates the total transaction cost, illustrated by the small and constant distance from the thick black line to zero.



hypothesis:

H0. The payoff of the box spread is less than the cost of the portfolio, i.e. (5) holds and so the options market is efficient;

H1. The payoff of the box spread exceeds the cost of the portfolio, i.e. (5) is violated and so the options market lacks efficiency.

We conclude this section with three remarks: i) Similar to the PCP test, passing the box-spread test strictly means that the market is efficient, otherwise we could conclude that the options market lacks efficiency; ii) Four simultaneously traded options are required in the box-spread test, while two options and one futures are in the PCP test. It is not obvious that the options market, tested via (5) is more efficient than the integrated option-futures market, tested via (4); iii) Both the PCP and box-spread tests screen for the frequency of violations, but they provide no information about the *magnitude* of arbitrage opportunities. This final comment is what motivates the next section.

3.3 Implied Transaction Costs

Recall that employing any of the empirical tests derived so far requires knowledge of the transaction costs. This section takes a different perspective. We explain how to 'back out' what we call 'implied' transaction costs by assuming validity of the PCP and box-spread tests. That is, we calibrate the implied transaction costs tc_o^* and tc_f^* from the boundary cases that pass the PCP and box-spread

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tests by solving the following simultaneous equations:

$$\begin{cases} 2\,tc_o^* + \dfrac{K}{F_{t,T}} tc_f^* &= \left| C_t(K,T) - P_t(K,T) - \left(1 - \dfrac{K}{F_{t,T}}\right) e^{-r(T-t)} \right|, \\ 2tc_o^* \left(K_1 + K_2\right) &= \left| K_2(C_{1t} - P_{1t}) - K_1(C_{2t} - P_{2t}) - e^{-r(T-t)} \left(K_2 - K_1\right) \right|. \end{cases}$$

The tests described in Sections 3.1 and 3.2 allow one to assess the *frequency* of arbitrage violations, whereas implied transaction costs reflect how severe the loss of market efficiency can become. Combining the violation frequencies and implied transaction costs together we may observe different periods during our sample, such as times when frequent arbitrage opportunities have low profits, or periods of infrequent but highly profitable arbitrage opportunities, or even frequent opportunities for highly-profitable arbitrages.

We take the following steps to calibrate tc_o^* and tc_f^*:

1. Noticing that the box-spread test only contains tc_o^*, we can estimate tc_o^* by solving just the second of the above equations. A box-spread violation occurs when:

$$x(tc_o) := 2tc_o \left(K_1 + K_2\right) - \left| K_2(C_{1t} - P_{1t}) - K_1(C_{2t} - P_{2t}) - e^{-rM_i} \left(K_2 - K_1\right) \geq 0 \right.$$

   Now suppose that, on day t, there are n box-spread violations, $x(tc_o, i)$, for $i = 1, \ldots n$. Then we obtain the implied cost tc_o^* for that day by minimizing the total squared arbitrage profit of all box-spread violation instances on that day, namely:

$$tc_o^*(t) = \arg\min_{tc_o} \sum_{i=1}^{n} x(tc_o, i)^2 \,. \tag{6a}$$

2. We estimate tc_f^* on day t using the PCP test with the previously estimated tc_o^*. That is:

$$tc_f^*(t) = \arg\min_{tc_f} \sum_{j=1}^{m} y(tc_f, j)^2 \,, \tag{6b}$$

   where j denotes the j-th PCP violation instance, $j \in [1, m]$, so that

$$y(tc_f, j) := \left| C_t(K,T) - P_T(K,T) - \left(1 - \frac{K}{F_{t,T}}\right) e^{-rM_j} \right| - 2\,tc_o^*(t) - \frac{K}{F_{t,T}} tc_f \geq 0 \,.$$

3. To alleviate any missing data problem at the daily frequency, we also construct a weekly and monthly series of implied transaction costs by taking the arithmetic average of the non-missing

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daily implied transaction costs within each week (Monday to Sunday) or month.

We note that this procedure makes the calibration of tc_o^* only relevant to the options market and independent of tc_f^*; but the calibration of tc_f^* depends on the value of tc_o^*. This is intuitive because tc_f^* relies on cross-market arbitrage portfolios whereas tc_o^* does not.

4 Data

For our investigation we use tick-by-tick options and perpetual futures data from the API of the Deribit platform as well as from the data vendor CoinMetrics.[19] We obtained the data on bitcoin trades from 1 January, 2019 to 31 September, 2023 and ether option trades from 1 April, 2021 to 31 September, 2023 (UTC time).[20] The data includes option prices, strikes, maturity time, and each trades timestamps and sizes.

4.1 Investigating Evidence of Wash Trading

There are some concerns about wash trading in cryptocurrency markets. For instance, Cong et al. (2023) found that wash trading on unregulated spot exchanges accounts for more than 70% of the reported volume, and that volumes on these exchanges display abnormal first significant digit distributions (known as Benford's law, see Benford, 1938), size rounding, and transaction tail distributions. Following their approach, we use Benford's law to investigate possible volume manipulations on the Deribit platform. For comparison purposes we also report Benford's law tests for trading volumes on S&P 500 index options, using data from 2019 to 2022.

The results are displayed in Figure 5. Looking at the perpetuals on the left hand side, we observe that the first digit distributions of the S&P 500 index option volume closely track the theoretical Benford's law distribution. The first digit distributions of BTC and ETH perpetuals volumes also resemble these of the S&P 500 index options volume. By contrast, regarding the options volumes on the right hand side, the first digit distributions of BTC and ETH option volumes indeed cast doubt on the absence of wash trading. Fortunately, the PCP and box-spread tests only use options

[19] CoinMetrics is a leading provider of crypto asset market and network data that provides access to historical and real-time data from over 30 of the world's leading spot and derivatives crypto exchanges. https://coinmetrics.io. Although the theoretical results are derived using calendar futures in the empirical work we use perpetuals because almost all the trading on their futures platform is on the perpetual, as noted already in the introduction. Because Deribit calendar futures are so illiquid and their prices follow the perpetual price very closely anyway.

[20] Unfortunately, there are plenty of missing data points from October 2023 onwards so we decided to end our sample here. We exclude data prior to 2019 because the trading volume was quite low.

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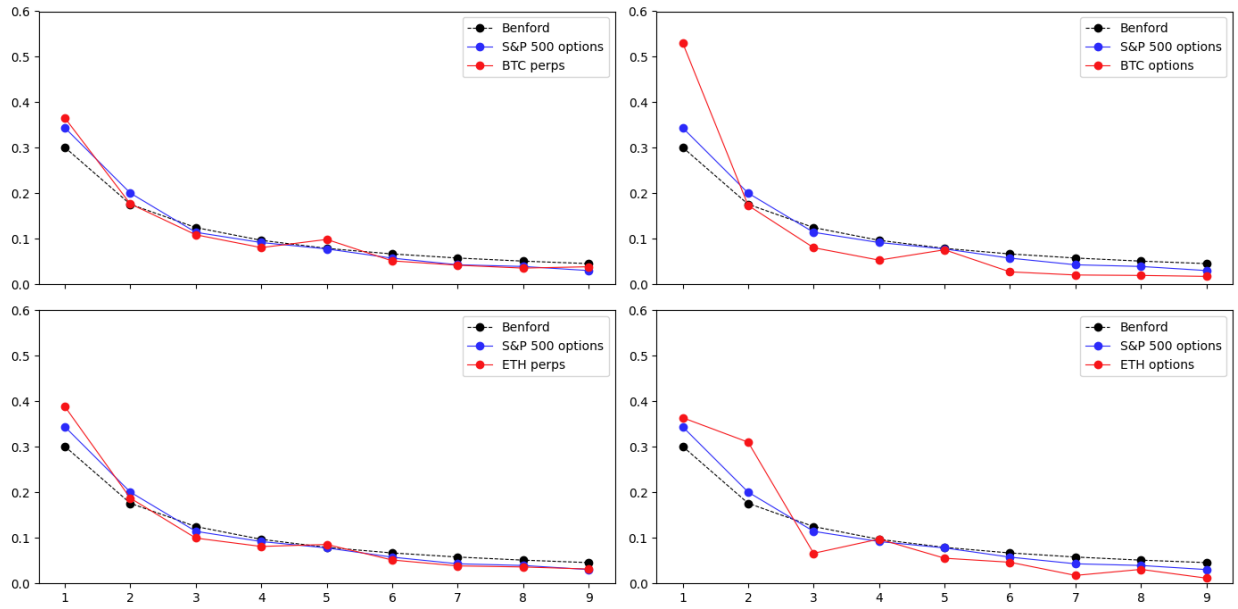
and perpetuals price data. We only use options volume to obtain approximations to the economic magnitude of arbitrage profit, see Section 5.3 below.

Figure 5: The First Digit Distribution of Benford's Law. The black line in each plot depicts the theoretic first digit distribution, known as Benford's law. The blue line depicts the first digit distribution for the S&P 500 volume from 2019 to 2022. The top panel also plots the first digit distribution for the BTC perpetuals and options trading volume from 2019 to 2023. The bottom panel also reports these of ETH perpetuals and options.



4.2 Analysis of Option Trade Data

Table 2 provides a detailed summary of the options trade data as well as a breakdown by option time to maturity. The summary in Table 2a shows a sharp growth in the number of trades and the number of traded contracts on bitcoin and ether options from 0.39 million trades (1.3 million contracts) in 2019 to 2.69 million trades (6.38 million contracts) in 2022. There are already 1.95 million trades (5.5 million contracts) in only the first three quarters of 2023. Similar observations can be made on ether options trading, from 1.34 million trades (25.98 million contracts) in three of the four quarters of 2021 in total, to 3.52 million trades (135.76 million contracts) in 2022. There are already 1.65 million trades (62.53 million contracts) in the first three quarters of 2023.

Table 2b shows that shorter maturity options are traded more, with the most liquid contracts being the shortest maturity options (less than seven days). The exception is ether options with maturity longer than 60 days. They are more often traded than those with a shorter time to maturity. Moreover, for both bitcoin and ether, call options are more actively traded relative to put options. The longer the option time to maturity, the more active call option trading is relative

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Table 2: Preliminary Statistics on the Bitcoin and Ether Option Trades Data. The sample is between 1 January, 2019 and 31 September, 2023 for bitcoin options, and between 1 April, 2021 and 31 September, 2023 for ether options. The table reports the number of trades and the volume of traded contracts (both in millions) of the entire data set of call (C) and put (P) options as well as per range of time-to-maturity (TtM, in days). The call/put ratio (C/P) reports the ratio of trades and trading volume on call options versus those on put options.

(a) Yearly sample on the number of trades and volume of options.

		Total		2019		2020		2021		2022		2023	
		Trade	Vol.	Trade	Vol.	Trade	Vol.	Trade	Vol.	Trade	Vol.	Trade	Vol.
Bitcoin	C & P	8.77	23.13	0.39	1.3	1.18	4.46	2.56	5.49	2.69	6.38	1.95	5.5
	C (in %)	5346	5519	5632	5383	5432	5194	5264	5452	5199	5218	5546	6228
Ether	C & P	6.52	224.26	0	0	0	0	1.34	25.98	3.52	135.76	1.65	62.53
	C (in %)	5424	5931	0	0	0	0	5300	5362	5429	5923	5517	6186

(b) Number and volume of trades by options' time to maturity (TtM).

TtM		Total		2019		2020		2021		2022		2023	
(in days)		Trade	Vol.	Trade	Vol.	Trade	Vol.	Trade	Vol.	Trade	Vol.	Trade	Vol.
						Panel A: Bitcoin							
≤ 7	C & P	4.58	10.86	0.25	0.75	0.81	2.95	1.22	2.52	1.19	2.48	1.12	2.17
	C (in %)	51.76	50.77	54.85	52.7	52.47	48.29	51.74	50.42	49.52	48.07	52.98	56.97
$(7, 15]$	C & P	1.4	4.12	0.06	0.21	0.14	0.62	0.44	1.03	0.45	1.22	0.32	1.03
	C (in %)	53.68	54.43	54.94	49.83	55.83	53.37	51.85	54.11	50.9	50.57	58.99	60.91
$(15, 30]$	C & P	1.04	3.08	0.03	0.13	0.11	0.47	0.33	0.73	0.34	0.93	0.23	0.83
	C (in %)	54.57	58.49	59.13	55.83	60.46	63.52	51.32	56.58	53.1	52.42	57.83	64.51
$(30, 60]$	C & P	0.83	2.53	0.03	0.11	0.06	0.24	0.27	0.57	0.32	0.83	0.16	0.77
	C (in %)	54.83	60.34	60.85	58.68	57.89	64.32	54.78	60.76	52.16	52.11	58.03	67.87
≥ 60	C & P	0.91	2.55	0.03	0.1	0.05	0.18	0.31	0.65	0.4	0.93	0.13	0.7
	C (in %)	59.1	66.1	64.7	63.07	61.83	60.22	56.88	63.33	59.52	65.09	60.89	71.91
						Panel B: Ether							
≤ 7	C & P	3.11	78.61					0.63	9.91	1.52	45.79	0.97	22.91
	C (in %)	51.99	50.84					52.14	49.84	50.55	49.19	54.18	54.56
$(7, 15]$	C & P	1.02	40.28					0.23	4.79	0.55	23.77	0.25	11.73
	C (in %)	53.21	54.43					51.75	51.15	53.18	53.79	54.6	57.06
$(15, 30]$	C & P	0.81	32.31					0.17	3.82	0.44	18.38	0.2	10.11
	C (in %)	54.41	59.68					53.05	53.7	54.27	57.9	55.92	65.17
$(30, 60]$	C & P	0.72	29.76					0.15	3.18	0.44	17.56	0.13	9.01
	C (in %)	57.57	65.45					54.36	57.02	58.56	63.76	57.86	71.73
≥ 60	C & P	0.86	43.3					0.17	4.27	0.57	30.26	0.11	8.77
	C (in %)	60.73	74.76					56.56	62.56	62.02	76.87	60.54	73.4

to put options. This may indicate an increasing proportion of arbitragers and long-term optimistic traders relative to hedgers in trading long-dated options.

For both bitcoin and ether, we set the crypto risk-free rate $r = 3$ bps per day which matches the offer rate from BitMEX. We also tested different borrowing and lending rates but our empirical results were not much affected. This is intuitive because liquid options/futures tend to have short maturities and so the impact of these rates is limited.

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4.3 Simultaneous Trades and the Matching Process

The perpetuals tick-by-tick data-set is huge, containing over 100 million entries. However, the data contains many trades which are identical in all dimensions (such as trade time and price) except for their size. We need to combine such trades into a single trade to shrink the size of the dataset enough to facilitate the matching process. It is important to note that this combining of identical trades does not affect the PCP and box-spread tests, because they do not use trading volume.

For the empirical tests listed in Section 3 we need to find matched assets as calls, puts and/or perpetuals that are traded simultaneously. We note that the efficiency tests require different groups of matched assets. For example the PCP test requires simultaneously traded futures, and puts and calls with the same strike price and maturity. But simultaneous trading across different contracts is highly unlikely in real data. In fact, relative to traditional markets where trading is concentrated within a limited time each day, because the Deribit platform runs continuously, the trades are relatively more spaced out which makes high-frequency matching even more difficult.

There is a trade off between the choice of sample-matching frequency and the number of matched pairs. Increasing the 15-second time window that we have selected would result in more matched pairs but could also cloud our conclusions with severe microstructure noise. Decreasing the window reduces noise but produces fewer matched pairs. Nevertheless our main conclusion seems to be robust to a wide range of time windows, such as 10-seconds and 30-seconds. We eventually settled on 15-seconds as the best choice – and this also follows Scholes and Williams (1977) and Lo and MacKinlay (1990).

4.4 Assumptions on Transaction Costs

The trading fees applied by the Deribit platform are reported in Table 1. In the efficiency tests we assume that each transaction is a taker order, thus accounting for a greater fee for placing the order. We assume portfolios are held to maturity, so the delivery fee for each contract also adds to the aggregate cost of executing our arbitrage strategies. From Table 1, the fees for a taker order are 5 bps for perpetuals and 3 bps for options, and the delivery fees are 25 bps for perpetuals and 15 bps for options. These two parts amount to 30 bps for perpetuals and 18 bps for options.

We use two scenario settings for transaction costs to estimate the magnitude of arbitrage profit in Section 5.2. Setting 1 assumes all taker orders can be filled immediately at the best price and setting 2 allows for slippage – we need to allow for this when the arbitrageur makes large orders in

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a market without sufficient liquidity at the top of the book for immediate execution of the order. Since slippage depends on the market liquidity, it would normally be greater for ether derivatives than for bitcoin derivatives. Also, the perpetual markets tend to have much tighter spreads and less slippage than the options markets. To keep round numbers, we make an ad-hoc adjustment to the taker order and delivery fees of setting 1, with an additional 10 bps for bitcoin perpetuals and 22 bps for bitcoin options and an additional 30 bps for ether perpetuals and 42 bps for ether options. This way we can state our two settings for transaction cost scenarios as follows:

Setting 1. The transaction costs are set as defined by the exchange without order slippage:

$$tc_f = 30 \text{ bps}, \quad tc_o = 18 \text{ bps} \quad \text{for both bitcoin and ether.} \tag{tc1}$$

Setting 2. We increase the transaction costs to allow for slippage in large orders:

$$tc_f = tc_o = 40 \text{ bps for bitcoin}, \qquad tc_f = tc_o = 60 \text{ bps for ether.} \tag{tc2}$$

5 Empirical Results

This section reports our empirical findings on the efficiency of the two options markets and their option-perpetual cross markets. The main takeaway is that the options market is more efficient than the option-perpetual cross market for both bitcoin and ether; and between the two, the bitcoin derivatives market is more efficient than the ether derivatives market. Overall, the efficiency of all derivatives are improving over time, although the bitcoin derivatives market lost some efficiency during some extreme events in 2021 and a similar, but less noticeable pattern happened in the ether derivatives market during early 2022.

Importantly, bitcoin and ether derivatives markets have their own characteristics. The bitcoin options-alone market is significantly more efficient than its option-perpetual cross market while the same cannot be said for ether. Moreover, we find several episodes of the derivatives' implied costs moving together with the underlying crypto prices. But it is the bitcoin perpetual implied costs and the ether options implied costs that move most closely with their underlying perpetual prices.

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5.1 Market Efficiency Tests with Scenario Transaction Costs

Table 3 reports our findings on the violation frequencies of each efficiency test derived in Section 3 for both the bitcoin and ether derivatives markets. The results are computed holding tc_o and tc_f at 40 bps per bitcoin contract and 60 bps per ether contract, namely using the setting tc2 which takes account of slippage costs for large orders.

There are four main takeaways from the results. First, a higher efficiency in the option market alone is observed in comparison to the option-perpetual cross market for both bitcoin and ether contracts. Additional results in Appendix B further echo this finding by showing that in the options market alone, the violation frequencies reduce very quickly over time, along with the transaction costs, indicating that the magnitude of arbitrage opportunities are not very large. However, the same cannot be said for the option-perpetual cross markets.

Second, the bitcoin derivatives market is generally more efficient than the ether derivatives market. The violation frequency in any year, for either test, is lower for bitcoin contracts than that for ether contracts. This is perhaps because the ether derivatives market is much younger than the bitcoin derivatives market, and consequently the market has had less time to evolve to a more efficient state.

Third, the efficiency of the derivatives markets are broadly improving over time, because we see a reduction in violation frequencies. The only exception is the bitcoin derivatives market in 2021 but not by a large amount. Interestingly, the price efficiency of longer-dated options (between 15 and 30 days) is improving over time to the extent that they become more price-efficient than the super-short-dated options of 7 days or less.

Fourth, the bitcoin and ether derivatives market have their own characteristics. For bitcoin the level of efficiency in the options market remains small and stable across our sample period, with violation frequencies being around 4% before 2022. By contrast, the integrated option-perpetual market is much less efficient until 2022 when the violation frequency of the PCP test suddenly drops to 0.78%. For the ether derivatives market, on the other hand, the options market and the option-perpetual cross market present similar levels of violation frequencies. The options market is marginally more efficient than the cross market. So it seems that ether derivatives markets might be attracting different types of traders and strategies compared with bitcoin. Both the ether options market and the option-perpetual cross market appear to have evolved rapidly to a more efficient state, with a violation frequency of only around 0.2% during the first nine months of 2023.

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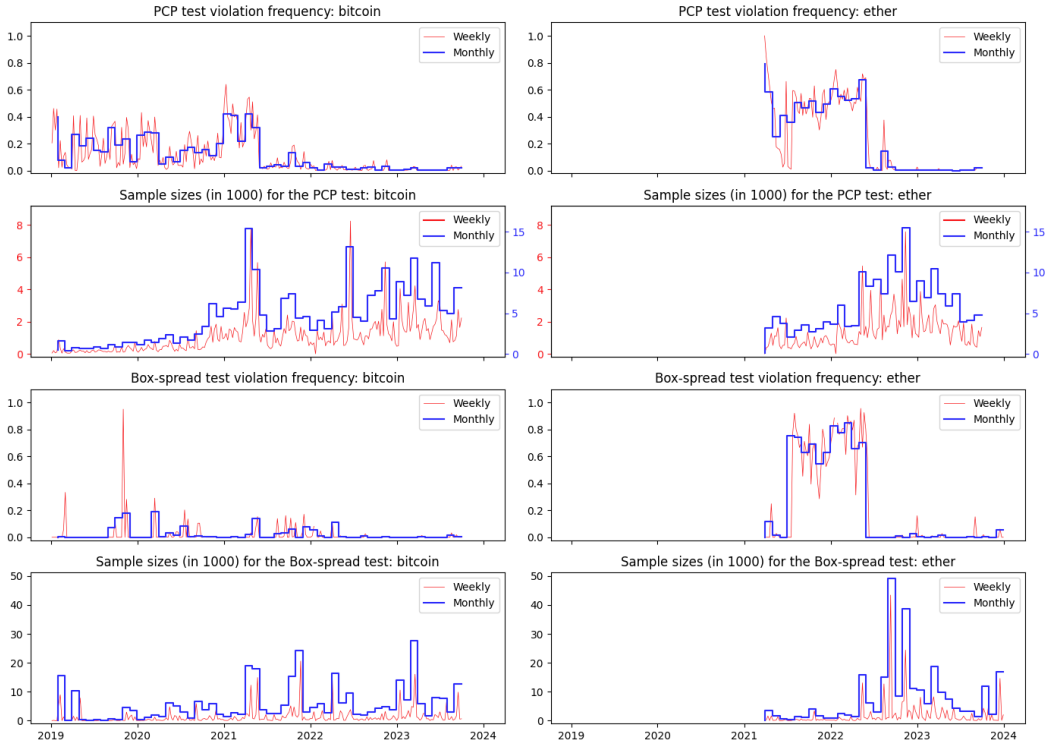
Table 3: Violation Frequencies (in %) of the Efficiency Tests. The sample is between 1 January, 2019 and 31 September, 2023 for bitcoin derivatives and between 1 April, 2021 and 31 September, 2023 for ether derivatives. For all results, transaction costs are set equally: 40 bps for bitcoin derivatives and 60 bps for ether derivatives. In Table (a) the sample size is reported in brackets.

(a) Yearly sample.

TtM	Bitcoin						Ether			
(in days)	2019	2020	2021	2022	2023	Total	2021	2022	2023	Total
Panel A:PCP test violation frequency (%)										
Full	18.07 (10831)	15.3 (30028)	22.89 (77090)	1.71 (72599)	0.78 (69963)	9.98 (260511)	38.45 (29746)	16.11 (95365)	0.22 (59773)	14.57 (184884)
≤ 7	9.94 (7455)	4.76 (22931)	3.3 (46934)	0.5 (49680)	0.12 (48794)	2.1 (175794)	25.15 (16979)	9.47 (64803)	0.08 (42455)	8.4 (124237)
$(7, 15]$	20.46 (2053)	27.45 (3002)	32.36 (13697)	2.42 (8765)	0.57 (10184)	15.77 (37701)	40.19 (4250)	24.11 (10256)	0.28 (7562)	19.04 (22068)
$(15, 30]$	25 (472)	51.11 (2342)	53.94 (6109)	0.86 (4315)	0.15 (5404)	24.97 (18642)	48.83 (2650)	30.08 (7725)	0.13 (4787)	23.9 (15162)
$(30, 60]$	69.69 (419)	83.55 (991)	75.9 (6082)	3.46 (6557)	3.37 (3684)	34.33 (17733)	51.57 (2616)	32.9 (6753)	0.12 (3296)	28.23 (12665)
> 60	89.35 (432)	85.96 (762)	87.91 (4268)	15.81 (3282)	15.87 (1897)	52.75 (10641)	86.62 (3251)	37.92 (5828)	4.18 (1673)	47.4 (10752)
Panel B:box-spread test violation frequency (%)										
Full	2.78 (36312)	2.48 (38187)	4.31 (99563)	3.39 (65195)	0.12 (88351)	2.61 (327608)	39.27 (15336)	9.64 (153852)	0.2 (64634)	8.98 (233822)
≤ 7	0.37 (17973)	2.7 (32856)	1.98 (62100)	0.72 (44871)	0.03 (49870)	1.22 (207670)	40.66 (8363)	6.6 (117761)	0.06 (36220)	6.9 (162344)
$(7, 15]$	0.35 (16779)	0.05 (3789)	5.46 (25153)	0.09 (5387)	0.44 (10601)	2.41 (61709)	35.57 (3652)	20.15 (9185)	1.3 (8211)	15.47 (21048)
$(15, 30]$	6.52 (322)	0.97 (1236)	4.32 (3471)	2.91 (1925)	0.07 (13533)	1.21 (20487)	19.32 (994)	24.41 (8121)	0.05 (6505)	13.94 (15620)
$(30, 60]$	6.92 (260)	10.9 (211)	29.84 (4386)	18.85 (9666)	0.25 (11973)	12.08 (26496)	44.75 (1448)	45.11 (4219)	0 (4669)	24.68 (10336)
> 60	86.61 (978)	23.16 (95)	4.99 (4453)	0.06 (3346)	0 (2374)	9.72 (11246)	54.95 (879)	9.1 (14566)	0 (9029)	7.39 (24474)

(b) Weekly and monthly time series.



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5.2 Implied Transaction Costs

Table 4a reports summary statistics of the implied transaction costs, tc_o^*, and tc_f^*, for bitcoin and ether. Recall that tc_o^* and tc_f^* are calibrated using the boundary cases of strict PCP and box-spread tests, assuming that the markets are free of arbitrage opportunities. Therefore a high implied transaction cost implies low market efficiency.

The univariate statistics of the implied costs reported in Panel A of Table 4a broadly confirm our findings from Table 3. First, the implied costs for bitcoin derivatives are always lower (in bps) than these of ether derivatives, indicating a higher efficiency for bitcoin. The results also show that this may not necessarily hold if counted in USD, because bitcoin is more expensive than ether.

Second, the implied costs for options are always smaller than those for the perpetual futures for both bitcoin and ether, indicating a higher efficiency in the options market alone than in the option-perpetual market, which is in line with our findings from Table 3. This holds for counting in both crypto and USD terms. Bitcoin and ether derivatives differ in that, for bitcoin contracts, the implied costs are significantly lower for options than the perpetual, while for ether contracts the implied cost for options and the perpetual are rather close. On the daily mean level, for instance, $tc_o^* = 5.14$ bps versus $tc_f^* = 37.96$ bps for bitcoin but $tc_o^* = 44.58$ bps versus $tc_f^* = 64.5$ bps for ether. Because professional arbitrage traders are able to use high levels of leverage to place large orders, we are not surprised that these average values for implied costs are close to the transaction cost scenario which accounts for slippage in large orders, namely the setting tc2 in Section 4.

In part 4b below the tabulated results we plot weekly and monthly time series of the implied transaction costs of bitcoin and ether options and futures. The results strengthen our findings about: (1) the implied cost of trading ether contracts is higher than trading bitcoin contracts; (2) the implied cost to trade bitcoin options is relatively stable and much smaller than trading bitcoin perpetuals; however, trading ether options and perpetuals incurs similar costs; (3) during 2022 there was a significant reduction in all implied costs which continues to the end of our sample. The plots also show that implied costs spike during some episodes, when bitcoin and ether are traded at very high prices. For instance, the spikes in the implied costs for bitcoin are largely reduced when changing the currency denominator from BTC to USD; similar observations can be made for ether derivatives during early 2022.

These implied cost plots highlight the times when there is a significant lack of efficiency in the derivatives markets, especially during 2021 for bitcoin and in early 2022 for ether. This echoes our

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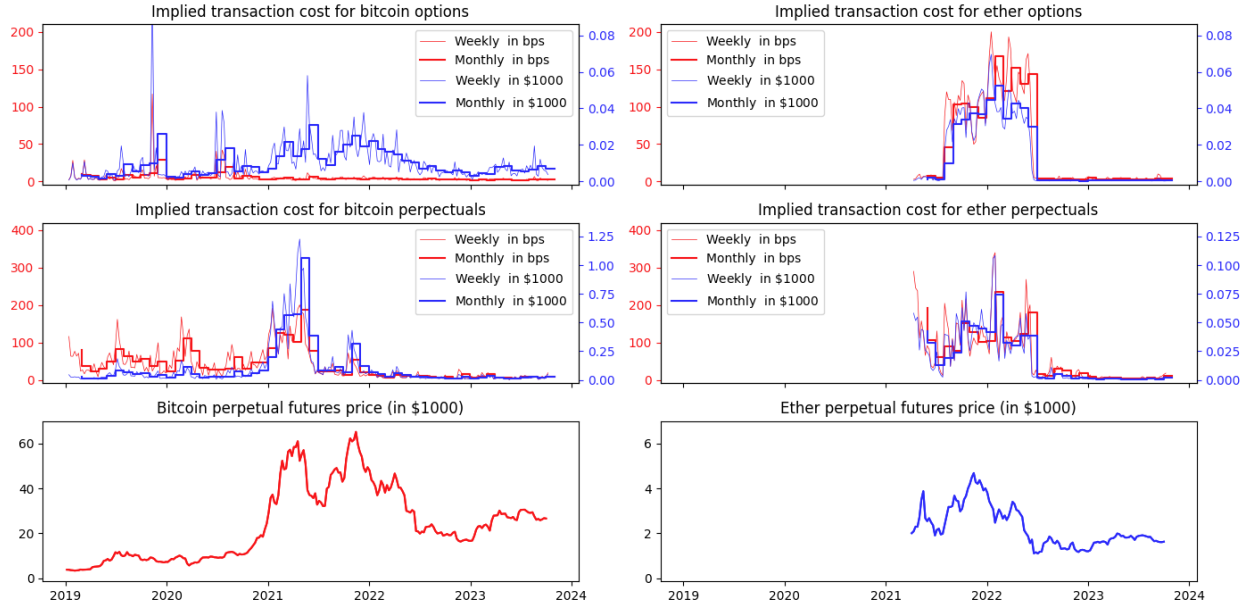
Table 4: Summary Statistics of the Implied Transaction Costs tc_o^* and tc_f^* for Bitcoin and Ether. The sample is between 1 January, 2019 and 31 September, 2023 for bitcoin derivatives and between 1 April, 2021 and 31 September, 2023 for ether derivatives.

(a) The univariate statistics of the implied transaction costs as well as their correlations with the first differences of the crypto prices (ΔP). The implied costs are computed and summarised at daily, weekly and monthly frequencies, denominated in bitcoin or ether, and in USD (in brackets).

	Bitcoin						Ether					
	Daily ($N = 1734$)		Weekly ($N = 248$)		Monthly ($N = 57$)		Daily ($N = 913$)		Weekly ($N = 131$)		Monthly ($N = 30$)	
	tc_o^*	tc_f^*	tc_o^*	tc_f^*	tc_o^*	tc_f^*	tc_o^*	tc_f^*	tc_o^*	tc_f^*	tc_o^*	tc_f^*
Panel A: Univariate statistics												
Min.	0.01 (0.01)	0.01 (0)	0.64 (0.62)	1.38 (3.9)	1.66 (1.54)	3.68 (8.62)	0.01 (0)	0.01 (0)	1.38 (0.22)	1.16 (0.24)	2.67 (0.36)	2.49 (0.46)
$Q1$	1.61 (2.51)	6.42 (12.27)	2.39 (3.95)	9 (18.06)	2.81 (4.83)	9.38 (20.95)	2.01 (0.33)	3.22 (0.56)	2.93 (0.43)	5.85 (0.9)	3.33 (0.56)	6.25 (1.05)
$Q2$	2.82 (5.2)	17.64 (28.71)	3.36 (6.13)	22.63 (33.16)	3.51 (7.73)	24.52 (35.8)	4.82 (0.86)	12.03 (1.84)	4.7 (0.83)	20.85 (3.15)	4.39 (0.69)	27.32 (4.53)
Mean	5.14 (9.73)	37.96 (97.37)	5.14 (9.72)	38.03 (97.3)	5.14 (9.74)	38.06 (97.81)	44.58 (13.58)	64.5 (18.52)	44.39 (13.53)	65.39 (18.68)	44.54 (13.57)	64.47 (18.51)
$Q3$	4.82 (10.71)	45.7 (72.2)	5.05 (11.89)	52.11 (74.1)	5.44 (14.02)	51.75 (73.46)	73.75 (23.75)	75.36 (21.91)	99.58 (30.77)	110.6 (33.9)	102.69 (34.09)	105.12 (38.58)
Max.	200 (293.98)	400 (2469.9)	89.16 (82.18)	209.8 (1293.43)	28.89 (30.79)	186.24 (1066.11)	200 (87)	400 (166.88)	200 (71.39)	383.74 (124.04)	167.86 (52.59)	235.62 (74.49)
S.D.	0.0013	0.0055	0.0008	0.0041	0.0005	0.0038	0.0066	0.0107	0.0060	0.0079	0.0058	0.0067
Skew.	11.06	3.02	6.90	1.82	3.35	1.70	1.40	2.03	1.04	1.52	0.87	0.86
Kurt.	153.73	14.67	64.96	6.20	16.33	6.11	3.50	6.17	2.59	5.04	2.08	2.70
Panel B: Implied transaction cost correlation matrix												
tc_f^*	−0.01 (−0.01)		−0.11 (0.12)		0.03 (0.09)		−0.07 (−0.07)		−0.03 (−0.18)		**0.73** (**0.66**)	
ΔP	−0.06 (−0.06)	0.02 (0.02)	0.03 (0.15)	**0.24** (**0.48**)	−0.03 (0.22)	**0.43** (**0.57**)	−0.08 (−0.08)	−0.07 (−0.07)	−0.22 (−0.02)	−0.01 (0.13)	0.16 (0.43)	−0.13 (0.12)

(b) Time series of the weekly and monthly implied transaction costs tc_o^* and tc_f^* for bitcoin (left panel) and ether derivatives (right panel) denominated in crypto or USD.



findings in Table 3a on the high violation frequencies of efficiency tests for bitcoin derivatives in 2021. Moreover, Panel B in Table 4a reports the correlations between the implied costs and the

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perpetual price. Those results add further evidence that bitcoin and ether derivatives markets have different characteristics. The panel shows that (1) bitcoin futures implied costs tc_f^* correlates highly with bitcoin futures price at both weekly and monthly level (0.24 and 0.43 in bps, and 0.48 and 0.57 in USD); indicating that the bitcoin perpetual implied costs (and level of market inefficiency) tend to move closely with the perpetual prices; and (2) the ether options implied costs tc_o^* is highly correlated with the ether perpetuals implied costs tc_f^* at the monthly level (0.73 in bps and 0.66 in USD) and also correlates with ether perpetuals in USD (0.43), indicating that the ether options implied costs (and level of market inefficiency) move closely with the ether perpetual prices. We provide further evidence in Section 5.4, to explore potential drivers of implied costs in bitcoin and ether markets.

5.3 Economic Magnitude of Arbitrage Opportunities

As a robustness check on our previous findings, we calculate the magnitude of each violation of the PCP and box-spread test boundaries, viz.

$$g_t^i(K,T) := \omega_i \cdot \left(\left| C_t(K,T) - P_t(K,T) - \left(1 - \frac{K}{F_{t,T}}\right) e^{-r(T-t)} \right| - 2\,tc_o - \frac{K}{F_{t,T}} tc_f \right)^+, \tag{8a}$$

$$h_t^j(K,T) := \omega_j \cdot \left(\left| K_2(C_{1t} - P_{1t}) - K_1(C_{2t} - P_{2t}) - e^{-r(T-t)}\,(K_2 - K_1) \right| - 2tc_o\,(K_1 + K_2) \right)^+. \tag{8b}$$

writing $C_t(K_1,T) := C_{1t}$, $P_t(K_1,T) := P_{1t}$, $C_t(K_2,T) := C_{2t}$ and $P_t(K_2,T) = P_{2t}$ for brevity and where i denotes the i-th instance of PCP violation, j denotes the j-th instance of the box-spread test violation, and ω_i and ω_j are the minimal trading volumes in the i-th and j-th matched pairs.

Table 5: The cumulative profit of arbitrage portfolios using bitcoin and ether derivatives. The sample is between 1 January, 2019 and 31 September, 2023 for bitcoin derivatives and between 1 April, 2021 and 31 September, 2023 for ether derivatives. The cumulative profit in USD is the sum of daily profit in bitcoin or ether, converted to USD using the close prices of bitcoin or ether in USD on each day that the profit is generated. In each case we report results under both scenario settings for transactions costs. That is tc1: $tc_f = 30$ bps, $tc_o = 18$ bps for both bitcoin and ether, and tc2: $tc_f = tc_o = 40$ bps for bitcoin, $tc_f = tc_o = 60$ bps for ether.

Bitcoin				Ether			
PCP		Box Spread		PCP		Box Spread	
tc1	tc2	tc1	tc2	tc1	tc2	tc1	tc2
697.85 BTC	507.70 BTC	58.02 BTC	23.88 BTC	16,931.00 ETH	13,134.92 ETH	9,298.24 ETH	8,100.88 ETH
$28,526,980	$22,368,323	$2,144,396	$879,424	$46,141,407	$36,825,746	$24,489,000	$21,258,286

Without knowing precise values for transaction costs tc_o and tc_f, we can only estimate arbitrage profits based on certain scenarios for these costs. Specifically, we use two settings described in

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Section 4.4, both of which include the Deribit order and delivery fees, and with setting tc2 adding an approximate cost of slippage for large orders. Next we sum the profits (8a) for the PCP arbitrage opportunities, each identified by a violation of the PCP test (4). Similarly we sum the profits (8b) for the box-spread arbitrage opportunities, each identified by a violation of the box-spread test (5). This way we obtain the overall profit generated from arbitrage opportunities, for each scenario setting, and we report the results in both token and USD in Table 5. We observe that:

1. The bitcoin arbitrage profits are accumulated between 1 January, 2019 and 31 September, 2023 and this is roughly twice as long as the period over which either arbitrage profits are accumulated, namely from 1 April, 2021 to 23 September, 2023. Nevertheless, the cumulative ether options market arbitrage profit is much greater than the cumulative bitcoin options profit. For instance, using the tc1 setting under the box-spread (option-market) strategy, we have a cumulative profit of \$24,489,000 (for ether) vs \$2,144,396 (for bitcoin), so bitcoin arbitrage profits are roughly 10% of ether arbitrage profits, even though the sample size for bitcoin is twice as long.

2. Cross-market arbitrage profits are greater, but in this case the cumulative ether profit is only about 60% greater than the cumulative bitcoin profit. For instance, using the tc1 (no slippage) setting under PCP we have \$46,141,407 (for ether) vs \$28,526,980 (for bitcoin).

3. The transaction cost scenario tc2 includes slippage from large trades, and this reduces profits relative to those under the no-slippage scenario tc1 by around 20% for both cross-markets, by about 15% for the ether options market and by about 60% for the bitcoin options market.

Now we consider the evolution of daily arbitrage profits. These are depicted as time series in Figure 6 using panels corresponding to the crypto (BTC left and ETH right) and for the transaction costs scenario settings tc1 (upper two rows) and tc2 (lower two rows). Note that we use the same log vertical scale for all eight combinations of coins, strategies and costs, for ease of comparison. In line with our discussion above, about cumulative profits, daily profits from the option market alone (box-spread strategy, shown in blue) are much smaller than the profits from the integrated option-perpetual market (PCP strategy, shown in red). It also shows how there have been sustained periods of profitable arbitrage during our sample. For instance, the left-hand red panels show that profitable bitcoin cross-market arbitrages existed throughout February 2020 and during the twelve-month period starting in July 2020. The left-hand panels confirm that the bitcoin options market (box-spread strategy, shown in blue) has fewer opportunities for arbitrage, especially under the

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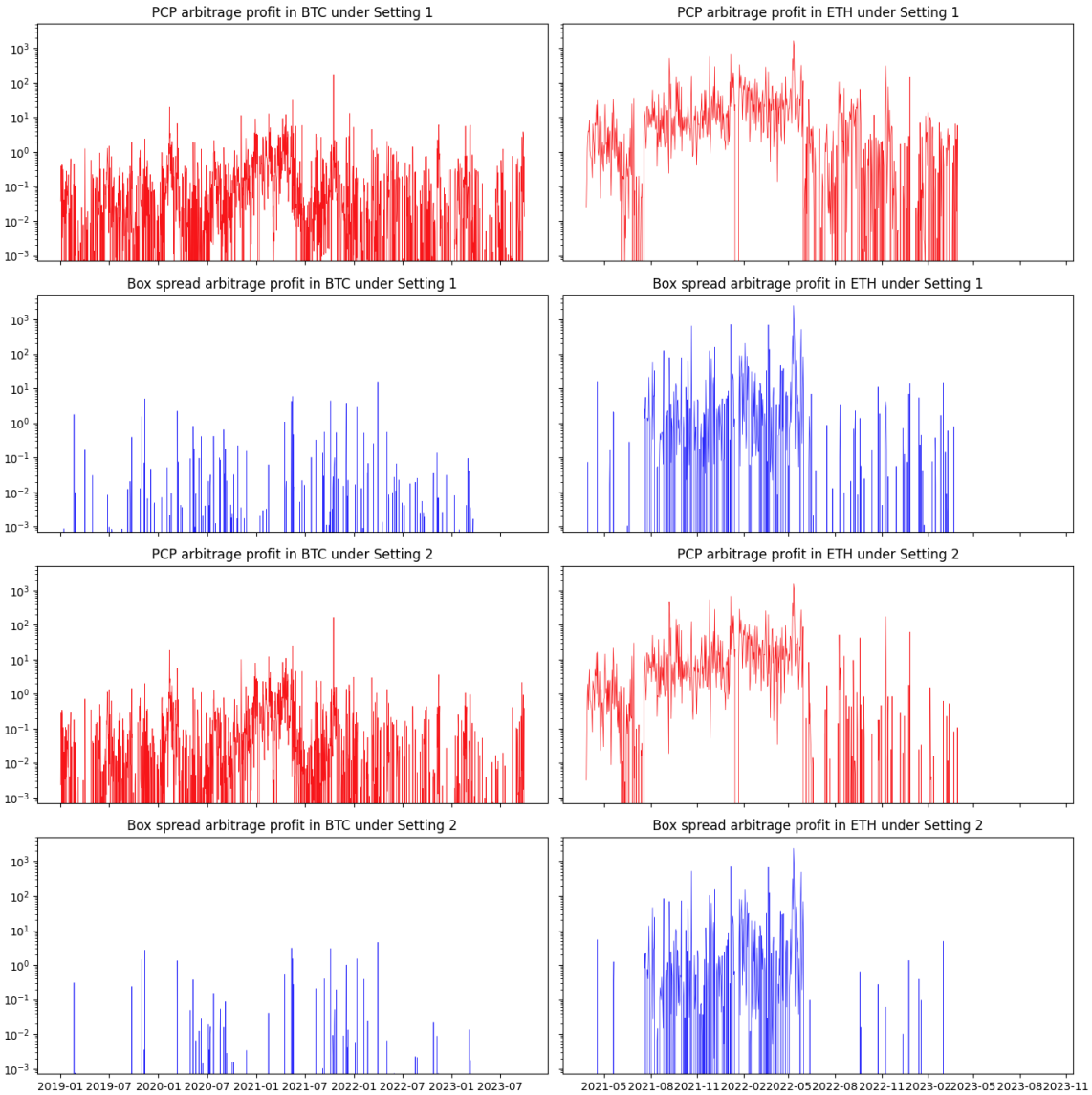
Figure 6: Arbitrage Profit in the Options and Option-Perpetual Markets. The figures report the daily arbitrage profit (in log scale): the crypto (BTC left and ETH right) and transaction cost assumptions are clarified above each figure, so is the arbitrage strategy (PCP in red and box spread in blue).



higher transaction costs assumption, the tc2 setting. The right-hand panels – and please note these cover a different sample to those on the left – tell a somewhat different story to bitcoin, except that there have also been sustained periods of profitable ether cross-market arbitrage just as in bitcoin. Indeed, the PCP strategy for ether was profitable on most days during the first year of the sample. However, by contrast with bitcoin, during the last year of the sample there was only one opportunity for profitable arbitrage which was on 11 November, 2022 when mainstream news about the collapse of the FTX exchange was first released.

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Table 6: Arbitrage Profit in the Options and Option-Perpetual Markets. The table reports the arbitrage profit (in 10,000 USD) generated during several major events; this represents the total of all profits and losses between 00:00:00 UTC to 23:59:59 on each of the major dates identified in column 1. The crypto (BTC left and ETH right) and transaction cost assumptions are clarified in the column headers. Most news items are retrieved from www.moneycontrol.com. The significantly large arbitrage profit (over 1 million USD) during any event is highlighted in blue.

	Bitcoin				Ether			
	PCP		Box spread		PCP		Box spread	
	tc1	tc2	tc1	tc2	tc1	tc2	tc1	tc2
Mass liquidation event (2021-04-18)	5.33	4.14	1.07	0.56	31.15	16.35	16.16	5.44
Binance platform outage (2021-05-19)	3.50	2.59	0.46	0.28	34.04	21.36	2.10	0.13
Visa joins NFT Frenzy (2021-08-27)	0.03	0.02	0	0	53.07	39.32	124.43	84.09
Ethereum blockchain glitch (2021-09-07/08)	0.69	0.36	0.13	0	714.74	679.32	79.86	70.06
SEC tweet about bitcoin futures ETF approval (2021-10-17)	177.38	166.71	0	0	4.87	3.04	0.06	0.03
Binance delegates 13M tokens to Uniswap DAO (2021-10-21)	1.48	0.89	0.10	0.05	161.75	126.70	645.80	520.23
A new strong COVID variant found (2021-11-26)	0.10	0.06	0	0	572.59	543.52	123.02	104.54
CME launches micro ether futures (2021-12-06)	0	0	0	0	295.63	282.88	157.46	153.15
FOMC minutes reveal FED's inflation concerns (2022-01-07)	0.17	0.13	0	0	703.95	681.58	713.03	702.35
ETH falls 27% in a week (2022-01-24)	0.12	0.04	0	0	335.61	288.11	89.44	80.85
ETH jumps 11% in a day (2022-02-04)	0	0	0	0	150.45	104.38	202.40	150.56
Ukraine legalizes cryptocurrencies (2022-03-17)	0.03	0.02	0	0	287.14	258.81	0	0
ETH test net merge success (2022-03-22)	0.72	0.58	0	0	210.12	199.99	694.98	675.21
ETH price rally (2022-03-24)	0.01	0.01	0	0	36.49	27.02	135.58	123.56
Terra/Luna attacks (2022-05-09∼13)	1.92	0.72	0.02	0	3699.95	3350.62	3940.29	3572.53
FSC hearing on cryptocurrency regulations (2022-05-26)	0.02	0	0.01	0	322.84	274.87	514.17	490.32
News of FTX (2022-11-09)	0.10	0.03	0	0	309.78	174.97	4.16	0.06
SEC cracking down on Crpto announced (2023-02-01)	0.34	0	0	0	13.93	0	0	0
SEC files 13 charges against Binance (2023-06-05)	0	0	0	0	0	0	0	0

Table 6 reports the 24-hr profit from arbitrage trading on the day of an important news event, using PCP and the box-spread results to pick out days when arbitrage is especially profitable due to higher than normal volatility. These news events are in two broad categories. The first type stems

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from investor's fear of loss in traditional financial markets, such as on 7 January, 2022 when the Federal Open Market Committee (FOMC) revealed major concerns by the Federal Reserve (FED) about rising inflation. On that day, speculation about a bear securities market drove up trading volumes on ether options, and the total arbitrage profit on ether options trades that calendar day reached roughly 700 ETH, whether using PCP or box-spread arbitrage, and even under the more conservative transaction cost setting tc2. The second type of arbitrage opportunity in ether options arises with investor's fears about crypto infrastructure, security or regulatory problems. These include the Terra/Luna attacks during 9-12 May, 2022, when the total arbitrage profits on ether options trades mounted up to over 3000 ETH in both the options market alone (box-spread arbitrage) and the option-perpetual market (PCP arbitrage). Over the two-day Ethereum blockchain glitch on 7-8 September 2021, the arbitrage profit on the option-perpetual market was also very high, exceeding 600 ETH. By comparison, the bitcoin market seems much more immune to major events. Only on 17 October 2021, the expectation of SEC approval of an ETF built on bitcoin futures generated a significant arbitrage profit of over 160 BTC.

5.4 Drivers of Arbitrage

Finally we attempt to shed light on the potential drivers of arbitrage profits by adapting the regression models of Crépellière et al. (2023), who examine the determinants of cross-exchange arbitrage in crypto spot markets.[21] We calibrate four regression models, with dependent variables being the implied transactions costs tc_o^* and tc_f^* for bitcoin and ether. Note that these implied transaction costs are inversely related to market efficiency, so the greater the implied transaction cost the higher the arbitrage profit. Arbitrage profits made from the options market alone are captured by tc_o^* whereas tc_f^* captures the profits made using cross-market arbitrage.

We construct several potential explanatory variables whose origin and/or construction is described in Appendix A along with their time-series plots, displayed in Figure A.1. These variables can be summarised into four different categories:

- Perpetual-specific (PS) features, for which we use the funding rate, FR and the daily trading volume of perpetuals, VLM_f.[22] Funding rates on perpetuals are positive during periods of

[21] We thank the editor (Prof. Clara Vega) and one anonymous referee for suggesting this.

[22] We removed the perpetual's realised volatility, VOL from regressions because of its high correlation with trading volume (0.66 for BTC and 0.71 for ETH) and Google search (0.74 for BTC and 0.56 for ETH). Trading volume is also highly collinear with realised volatility (0.41 for BTC and 0.65 for ETH) and so we also removed this variable from the regressions, to avoid problems with multicollinearity.

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contango and only go negative during periods of backwardation market. They also increase during bull markets and decrease during bear markets (He et al., 2022; Angeris et al., 2023). Hence, a significant positive (negative) coefficient on FR indicates that arbitrage profits tend to be greater (smaller) during bull markets than during bear markets. A significant positive (negative) coefficient on VLM_f indicates that arbitrage profits tend to be greater (smaller) during periods of high trading volume, and when the market is particularly volatile;

- Option-specific (OS) features include the put-call ratio, PCR namely the ratio of daily trading volumes on puts relative to calls, and the daily trading volume on all options, VLM_o. The put-call ratio gauges the general mood of informed traders in the underlying market. For equity options it is usually greater than one, and increases when investors fear a downward price jump so they buy more puts to hedge downside risk. Upward price jumps rarely happen in stock prices or indices, but they are very common in bitcoin and ether. Thus, the put-call ratio for bitcoin and ether options tends to be smaller than for equity options, as can be seen from Figure A.1. If arbitrage is more (less) profitable when traders fear a crash in bitcoin/ether prices then the coefficient for PCR will be significantly positive (negative). Interestingly, trading volumes on perpetuals and options are weakly inversely related,[23] so we anticipate the coefficients on VLM_o may have opposite signs to those for VLM_f;

- Blockchain-specific (BS) characteristics are captured by two variables: ACT, which for bitcoin is the average cost per transaction and, for ether, transaction costs are proxied by the gas fee; and network activity, NET which is the hash rate for bitcoin and unique addresses for ether. The value of ACT increases when blockchain traffic is congested and transactions are slowed down, making the crypto markets less efficient. Therefore we expect a positive coefficient on ACT. Because network activity increases with price volatility we expect NET to have a coefficient of the same sign to VLM_f;[24]

- Market-wide (MW) conditions are captured by the S&P 500 variance risk premium, VRP and the Aruoba et al. (2009) Business Condition Index, ADS.[25] Both variables could affect investors' attention to crypto markets. Market-wide risk aversion, proxied by the S&P 500

[23] The sample correlation is -0.03 for BTC and -0.18 for ETH.

[24] However, we do not remove NET variable because the sample correlation of NET with VLM_f is only 0.21 for BTC and 0.01 for ETH, where the multicollinearity problem is not of concern.

[25] Following Yao et al. (2021) we also (initially) included Google search activity for bitcoin and/or ether, GS. However, as expected, GS has uncomfortably high correlations with VLM_f (0.89 for BTC and 0.78 for ETH) and so we omitted GS from the final regressions.

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variance risk premium, may affect arbitrage in crypto markets positively or negatively. On the one hand, higher risk aversion in stock markets could lead to greater inflows to crypto markets, making them more efficient and therefore lowering arbitrage profits; on the other hand, larger risk aversion globally would, ceteris paribus, decrease the tendency for professional traders to take on additional risks in crypto. Macro-economic outlook, proxied by the ADS Index, may have a negative correlation with order flows in crypto – the worse the economic outlook the more investors seek sources of diversification from stock markets. If this hypothesis is true then the ADS index should be negatively correlated with crypto market efficiency and therefore positively correlated with implied transaction costs.

Before presenting the results of our regressions we make some remarks. First, from the initial set of potential explanatory variables described in Appendix A, the final regressions contain only eight explanatory variables because the high pair-wise sample correlations between them revealed several potential variables that might capture the same effect, and multicollinearity would be an issue if they were included in the regressions. The final regressions are the result of a testing-down process during which the signs and significance of the remaining coefficients were hardly effected (results being available from the authors on request). However, serially correlated errors were an issue so we used generalised instead of ordinary least squares to estimate each model. Secondly, for some variables, the results for ether may differ from those for bitcoin, because investors may be driven by different motivations. Bitcoin is a more speculative asset, which attracts manipulative professional traders, whereas ether has more commodity and equity value (being the native token of a smart-contract blockchain) and so might attract more fundamental investors. Finally, arbitrage driving variables in options market alone may also differ from those that drive cross-market arbitrage (namely the possibility for different types of traders or investors in these markets).

Table 7 reports the final results, using the same explanatory variables for all four regressions. Most variables are highly significant determinants of each kind of arbitrage although, as previously discussed, we do find differences between the drivers of profits in the four different markets. Variables that significantly increase arbitrage profits in all markets are: VLM_f, the trading volume of perpetuals (which is highly and positively correlated with realised volatility and investor attention via Google search); ACT, the blockchain average transaction costs; and NET, the blockchain network activity. With a very highly significant coefficient of 0.74, bitcoin perpetuals trading volume is by far the most important driver in bitcoin options. But for ether options, the activity on the Ethereum blockchain, NET is the most powerful arbitrage driver with a coefficient 0.63. Cross-

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Table 7: Determinants of Arbitrage Profits. Generalised least squares estimates of linear regressions on four categories of explanatory variables with perpetual-specific (PS), option-specific (OS), blockchain-specific (BS) and market-wide (MW) features. Data are daily between 1 April, 2021 and 31 September, 2023. The dependent variables are implied transactions costs: tc_o^* for arbitrage in the option market alone, and tc_f^* for arbitrage in the option-perpetual cross market. All variables are described in the main text and plotted in Appendix A. They are then standardised to have zero mean and unit variance so that the size as well as the significance of coefficients have meaning. ***, **, * denote 1%, 5% and 10% significance levels.

		Option Market Alone		Option-Perpetual Cross Market	
		BTC	ETH	BTC	ETH
PS	FR	-0.19***	0.06*	0.15**	0.08
	VLM_f	0.74***	0.22***	0.21***	0.29***
OS	PCR	-0.17***	0.10***	0.23***	0.13***
	VLM_o	0.00	-0.15***	0.15***	-0.10***
BS	ACT	-0.03	0.25***	0.19***	0.36***
	NET	0.10**	0.63***	0.33***	0.19***
MW	VRP	0.02	0.13***	-0.02	0.06
	ADS	-0.12***	-0.17***	0.27***	-0.06
	Const.	-0.00	-0.01	0.03	0.00
	Adj. R^2	0.43	0.76	0.60	0.54

market arbitrage determinants are more diverse, with VLM_f, ACT and NET having similar and very highly significant effects.

The funding rate, FR and especially the put-call ratio, PCR are also important determinants of arbitrage profits, in both the cross market and the options market alone. Note that FR and PCR have negative signs for the bitcoin options market, but positive signs in the other markets. In other words, bitcoin option arbitrage is more profitable in bitcoin bear markets, or when the put-call ratio decreases, whereas arbitrage in the other three markets is more profitable when prices are increasing or when puts become more heavily traded relative to calls. One can explain this by noting how crypto investor fears, which are reflected by an increased PCR, tend to increase when prices gains become too rapid and so, if history repeats itself, yet another crash is imminent.

The variance risk premium, VRP only drives up arbitrage profits in the ether options market, not in the other markets. We had no prior on the sign on VRP because higher risk aversion in stock markets could attract more less-informed, retail investors to crypto, making arbitrage opportunities more profitable via the high-frequency trading algorithms that the professional traders employ. Probably there are already plenty of informed bitcoin option traders,[26] which verifies our findings in the previous section, that arbitrage has been much more profitable in ether derivatives

[26]For instance, Alexander et al. (2023b) find evidence of both volatility-informed and directional-informed traders.

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than in bitcoin derivatives. Finally, profits from options arbitrage tend to be lower when general business conditions (captured by the ADS index) improve. This could be due to professional traders preferring to trade traditional options during good market conditions, rather than to deploy their algorithms in the much more risky crypto markets.

6 Conclusions

We first revise a long list of efficiency tests proposed in the literature given that the crypto derivatives are settled and quoted in crypto instead of fiat-currency. Using these revised tests and tick-by-tick data on bitcoin and ether options as well as minute-level data on perpetual futures, we find that both bitcoin and ether options and option-perpetual markets are evolving towards a more efficient state, which supports the findings of Alexander et al. (2023b). Moreover, we show that the options market alone is more efficient than the option-perpetual cross market for both bitcoin and ether. For both, the price efficiency of long dated options has significantly improved over time and the bitcoin derivatives markets are more efficient than the ether markets.

We have measured the size of arbitrage profits obtained by following our options-only strategies (executed on violation of the box-spread test) and our options-futures strategies (executed on violation of the PCP test) under two realistic scenarios for transactions costs, one which adds an approximate cost for slippage in large orders and so is more conservative than the other. We emphasize that our findings are specific to bitcoin and ether derivatives traded on the Deribit platform. Particularly, the transaction costs may not necessary reflect the transaction costs applied in other crypto derivatives exchanges. But given the hugely dominant market share of the Deribit options exchange, we believe our results are robust.

Over the entire sample, the bitcoin arbitrage profits are accumulated between 1 January, 2019 and 31 September, 2023 which is roughly double the period over which ether arbitrage profits are accumulated, namely from 1 April, 2021 to 23 September, 2023. Nevertheless, the cumulative ether market arbitrage profit is much greater than the cumulative bitcoin profit. For instance, under the more conservative transaction cost scenario the PCP cross-market arbitrage strategies generate profits of around 500 BTC (20 million USD) and 13,000 ETH (36 million USD).

Our investigation also offers several interesting observations about the determinants of arbitrage in bitcoin and ether derivatives markets. After an extensive empirical investigation, we discover that the trading volume on perpetuals (which is highly correlated with both volatility and retail

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investor's sentiment proxied via Google search) is the most important driver of arbitrage in crypto derivatives markets overall, especially in the bitcoin options market. Our explanation is that, due to the complete lack of regulation in these markets, it is the prevalence of algorithmic traders that use malpractice (such as spoofing) to create arbitrage opportunities, especially during periods of high volatility, of which they themselves may in fact be the cause.

Also of importance are blockchain specific factors, particularly the activity in the network and the cost of blockchain transactions. When either of these is high, arbitrage becomes more profitable. We suggest that such profits may also be made at the expense of the less-informed, retail traders, but the mechanism at play here would be different. The results in Cheng et al. (2021) suggest that retail traders tend to select small margins, much smaller than the risk-based margins required on regulated derivatives exchanges like the CME. Also, like all other unregulated crypto exchanges, Deribit makes no margin calls and so retail traders run a high risk of their entire account and all their positions on Deribit being automatically liquidated, without notice, unless they can top up their collateral account very quickly. When the blockchain network is congested and costly, it becomes more difficult to add collateral via Deribit's hot wallet. We suggest that this is the reason why network activity is another main driver of arbitrage profits on Deribit platform.

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Appendix

A Explanatory Variables for Regression Models

First we obtain several variables which capture the characteristics of option and perpetual futures markets – the put-call ratio and daily trading volume on options, and the funding rate and daily trading volume for the perpetual contract.[27] In addition, arbitrage opportunities might arise more frequently when markets are more volatile and to capture this we use an historical volatility computed via an exponentially-weighted moving average of squared log returns which is then square-rooted and annualised using 365 days per year. Daily data on all these variables were obtained from the CoinMetrics API.

In the blockchain specific category, to capture on-chain trading costs we use the gas price for Ethereum and the average cost per transaction for Bitcoin, which is calculated by taking the total revenue paid to all block producers (including both transaction fees and block rewards) and dividing this total by the number of transactions on that day. To capture the extent of network activity we use the hash rate for bitcoin and the number of unique addresses for ether. For ether, the gas prices and cumulative unique addresses, and for bitcoin the average cost per transaction and network hash rate are all downloaded from YCharts.

As indicators of an investor's general attitude to traditional and crypto markets conditions, we compute the S&P 500 variance risk premium using the definition in Bakshi and Madan (2006) also in Bollerslev et al. (2011). This requires daily close prices of the S&P 500 index and the VIX implied volatility index (both sourced from Yahoo Finance) and daily values of S&P 500 realised variance (sourced from the Oxford-Man institute). We also used search counts for the terms 'bitcoin' and 'ether' (sourced from Google Trends), and the Aruoba et al. (2009) Business Condition Index (obtained from the Federal Reserve Philadelphia) which summarises the information of six widely-followed macroeconomic series. As the index increases or decreases general business conditions improve or deteriorate.

[27] Periodic funding payments tie the perpetual market price to a spot index price: when the funding rate is positive, long positions pay to short positions, and vice versa when the funding rate is negative. The sign and size of the funding rate reflects the bull and bear markets conditions, where the futures curve tends to be in backwardation and contango respectively.

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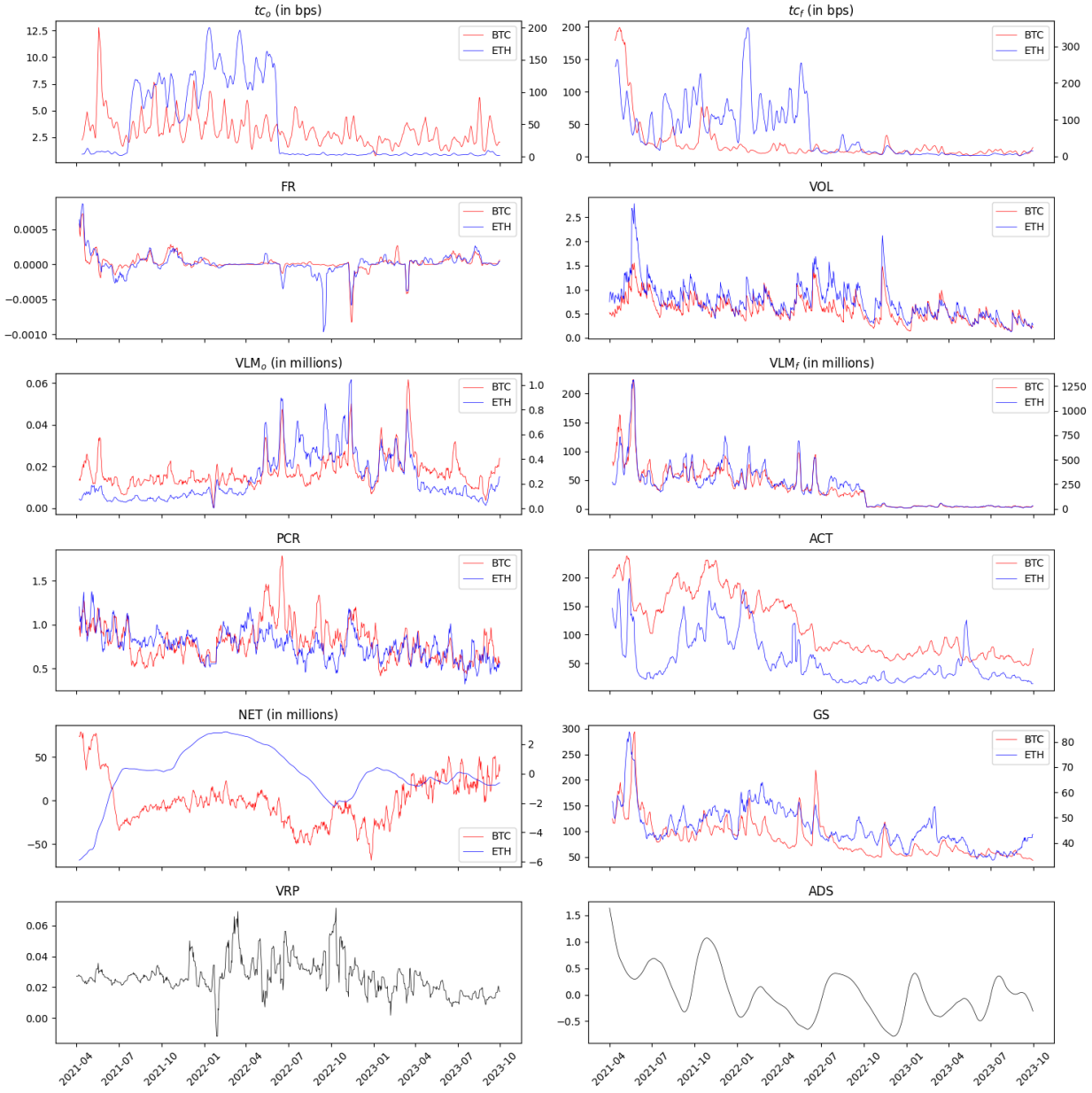
Figure A.1: Variables for Regression Analysis Time series of the implied transaction costs in both bitcoin and ether perpetuals and options markets (tc_o and tc_f) and all the explanatory variables used in the regression models. For tc_o, tc_f, VLM_o, VLM_f, NET and GS, the time series for ETH are plotted against the secondary Y-axis for better display.



To summarise, the explanatory variables include the funding rate (FR), historical volatility (VOL), volume in both options and perpetuals markets (VLM_o and VLM_f), the put-call ratio (PCR), the average cost per transaction for bitcoin or the gas fee for ether (ACT), the size of activity in the network (NET), google search counts on bitcoin and ether (GS), the S&P 500 variance risk premium (VRP) and the business condition index (ADS). Because the series for NET

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show clearly increasing trend, in our regression models we use the residuals from a regression on a time trend. All other variables are already stationary but many of them are excessively fluctuating which make them unsuitable for regression analysis; therefore the dependent variables tc_o and tc_f are smoothed using the past 7-day averages, and so also are the explanatory variables FR, VLM_o, VLM_f, PCR, ACT and GS. The results are displayed in Figure A.1.

B Weak Arbitrage Boundaries and Efficiency Tests

This section derives additional arbitrage boundary tests derived from the fiat-currency-free PCP in Section 3, following Ackert and Tian (2001). The PCP and box-spread tests reported in Section 3 are strict tests but as reformulations of the PCP, the boundary tests derived in this section are relatively loose. Failing any of these tests presents a lack of market efficiency with potential existence of arbitrage opportunities. But passing any of these tests does not necessarily imply that the market is efficient. These comments will be clarified in the following as we explain of how these tests are derived.

B.1 Arbitrage Boundaries in the Option-Futures Cross Market

We derive the arbitrage price boundaries outside of which the cryptocurrency call or put prices present arbitrage opportunities and thus imply that the market is inefficient. As prices of put and call options never go negative, the two parities also yield boundary constraints for put and call options separately. For instance, the PCP implies:

$$C_t(K,T) = P_t(K,T) + \left(1 - \frac{K}{F_{t,T}}\right) e^{-r(T-t)} \geq \left(1 - \frac{K}{F_{t,T}}\right) e^{-r(T-t)}.$$

Incorporating transaction costs, we have

$$C_t(K,T) \geq \left(\left(1 - \frac{K}{F_{t,T}}\right) e^{-r(T-t)}\right)^+ - tc_o - \frac{K}{F_{t,T}} tc_f\,. \qquad \text{[Call boundary test]} \quad \text{(B.1a)}$$

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Otherwise, one can write the undervalued call and take proper short position using futures. In parallel, for put options, we have:

$$P_t(K,T) \geq \left(\left(\frac{K}{F_{t,T}} - 1\right) e^{-r(T-t)}\right)^+ - tc_o - \frac{K}{F_{t,T}} tc_f \,. \qquad \text{[Put boundary test]} \tag{B.1b}$$

B.2 Arbitrage Boundaries in the Options Market

(1) Call and Put Spreads

Unlike the box spread consisting of both call and put options, the call (put) spreads involve only two call (put) options with different strikes and same maturity. Taking the call spread for example, at time maturity T, the terminal payoffs of two types of call options C_{1t}, C_{2t} with characteristics (T, K_1) and (T, K_2), $K_1 < K_2$, will take the following form:

$$K_1\, Q(C_{2t}) - K_2\, Q(C_{1t}) = \begin{cases} K_1 - K_2, & \text{if} \quad S_T \geq K_2 > K_1\,, \\ K_1 - \frac{K_1 K_2}{S_T} \geq K_1 - K_2, & \text{if} \quad K_2 > S_T \geq K_1\,, \\ 0 > K_1 - K_2, & \text{if} \quad K_2 > K_1 > S_T\,. \end{cases}$$

where $Q(C)$ denotes the payoff of option C. That means we always have

$$K_1\, C_{2t} - K_2\, C_{1t} \geq e^{-r(T-t)}\,(K_1 - K_2)\ .$$

When transaction costs are involved, the left-hand side can be negative but its absolute value would not exceed the cost of constructing the spread $tc_o(1/K_1 + 1/K_2)$. Therefore, we have the following call spread boundary:

$$K_1\, C_{2t} - K_2\, C_{1t} \geq e^{-r(T-t)}\,(K_1 - K_2) - tc_o\,(K_1 + K_2)\ . \quad \text{[Call spread test]} \tag{B.2a}$$

Similar arguments yield the corresponding put spread:

$$K_2\, P_{1t} - K_1\, P_{2t} \geq e^{-r(T-t)}\,(K_1 - K_2) - tc_o\,(K_1 + K_2)\ . \quad \text{[Put spread test]} \tag{B.2b}$$

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(2) Butterfly Spreads

The butterfly call (put) spread involves three call (put) options with identical maturity but different strike prices. Without loss of generality, let us assume that $K_1 < K_2 < K_3$. The spread is constructed by writing options with K_2 and buying options with K_1 and K_3. Let us set

$$\omega = \frac{K_1(K_3 - K_2)}{K_2(K_3 - K_1)} > 0.$$

At time maturity T, the butterfly call spread constructed using call options C_i, $i = 1, 2, 3$, with characteristics (T, K_i), will have the following payoff:

$$\omega K_2 K_3\, Q(C_{1t}) - K_1 K_3\, Q(C_{2t}) + (1-\omega) K_1 K_2\, Q(C_{3t}) = \begin{cases} 0, & \text{if } S_T \geq K_3 > K_2 > K_1\,, \\ \left(\frac{K_1 K_2 K_3}{S_T} - K_1 K_2\right)(1-\omega) \geq 0, & \text{if } K_3 > S_T \geq K_2 > K_1\,, \\ \omega\left(K_2 K_3 - \frac{K_1 K_2 K_3}{S_T}\right) \geq 0, & \text{if } K_3 > K_2 > S_T \geq K_1\,, \\ 0 & \text{if } K_3 > K_2 > K_1 > S_T\,. \end{cases}$$

That means the spread must satisfy the following inequality:

$$\omega K_2 K_3\, C_{1t} - K_1 K_3 C_{2t} + (1-\omega) K_1 K_2\, C_{3t} \geq 0\,.$$

When transaction costs are considered, the left-hand side can be negative but its absolute value would not exceed the cost of constructing the spread:

$$tc_o\,(\omega K_2 K_3 + K_1 K_3 + (1-\omega) K_1 K_2) = 2tc_o K_1 K_3.$$

Therefore, we have the following butterfly call spread boundary:

$$\omega K_2 K_3\, C_{1t} - K_1 K_3 C_{2t} + (1-\omega) K_1 K_2 C_{3t} \geq -2tc_o K_1 K_3\,. \quad \text{[Butterfly call spread test]} \tag{B.3a}$$

Similarly, the butterfly put spread boundary is

$$\omega K_2 K_3\, P_{1t} - K_1 K_3 P_{2t} + (1-\omega) K_1 K_2 P_{3t} \geq -2tc_o K_1 K_3\,. \quad \text{[Butterfly put spread test]} \tag{B.3b}$$

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B.3 Empirical Results on Weak Arbitrage Opportunities

Table B.1 reports the full sample results of the violation frequencies of the above mentioned arbitrage boundaries as well as those of the PCP and box-spread tests explained in the main text as a benchmark, based on different transaction cost assumptions. We make several observations. First, as expected, the higher the aggregate transaction costs assumed, the lower the violation frequencies. Second, we observe much lower violations of all loose boundary tests relative to those of the strict tests (*i.e.* PCP and box-spread tests), regardless of our transaction cost assumptions. This is intuitive because by design, passing the loose tests does not necessarily suggest that the market is efficient. Third, for the loose tests, call option based violations are less than put option based violations. This is likely because call options are more liquid (with larger trading volume) than puts, as shown in the summary Table 2.

Table B.1: Violation Frequencies (in %) of the Efficiency Tests within the Full Sample. The sample is between 1 January, 2019 and 31 September, 2023 for bitcoin derivatives and between 1 April, 2021 and 31 September, 2023 for ether derivatives. For all results, tc_f are set at 40 bps for bitcoin perpetuals and 60 bps for ether perpetuals, which are roughly the averages of full sample implied costs respectively.

		Bitcoin					Ether				
		Nobs.	tc_o (in bps)				Nobs.	tc_o (in bps)			
			0	10	**40**	80		0	40	**60**	120
Panel A: Option-Perpetual Markets											
Boundary	**PCP**	260,511	24.74	17.86	9.98	5.85	184,884	26.83	17.57	14.57	9.02
	C	4,494,261	0.04	0.03	0.01	0.01	3,192,453	0.95	0.81	0.75	0.62
	P	3,925,373	0.13	0.12	0.09	0.07	2,704,387	1.51	1.36	1.29	1.13
Panel B: Options Market-Alone											
Box Spread		327,608	100	16	2.61	0.5	233,822	100	11.03	8.98	4.69
Spread	C	1,710,171	0.43	0.15	0.03	0.01	1,571,278	0.95	0.64	0.57	0.44
	P	1,597,249	0.86	0.56	0.28	0.13	1,228,555	1.23	0.75	0.67	0.5
Butterfly	C	3,542,688	3.95	1.31	0.45	0.13	3,639,314	12.56	5.54	4.48	2.61
	P	3,529,218	7.49	4.65	2.71	1.74	2,372,898	13.64	6.7	5.81	4.17

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